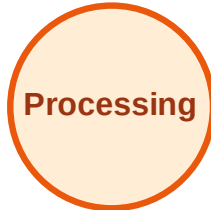


BFS Traversal

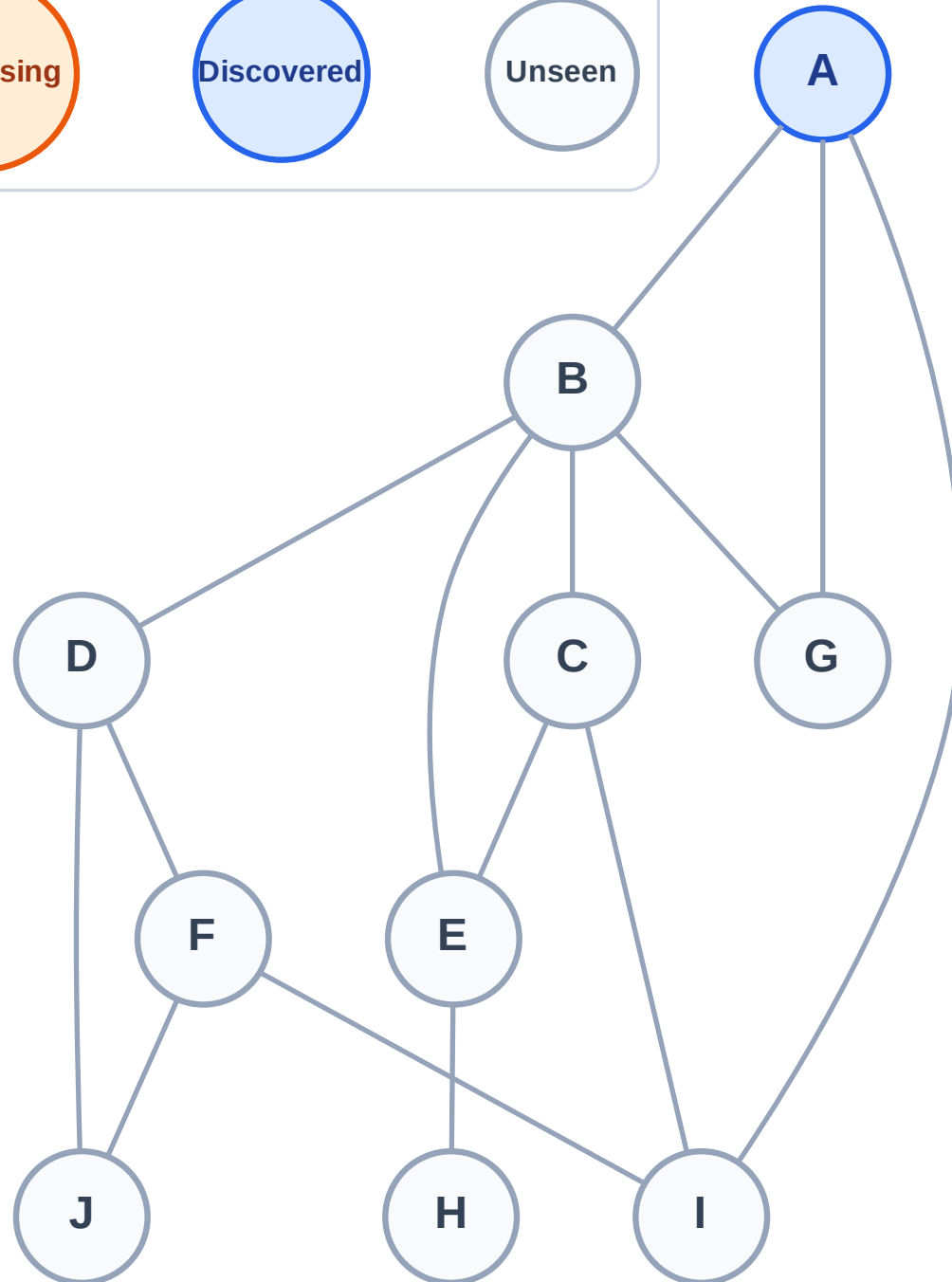
Step 1: Start at A

Legend



Queue: A

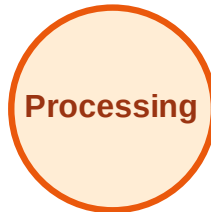
Visited: (none)



BFS Traversal

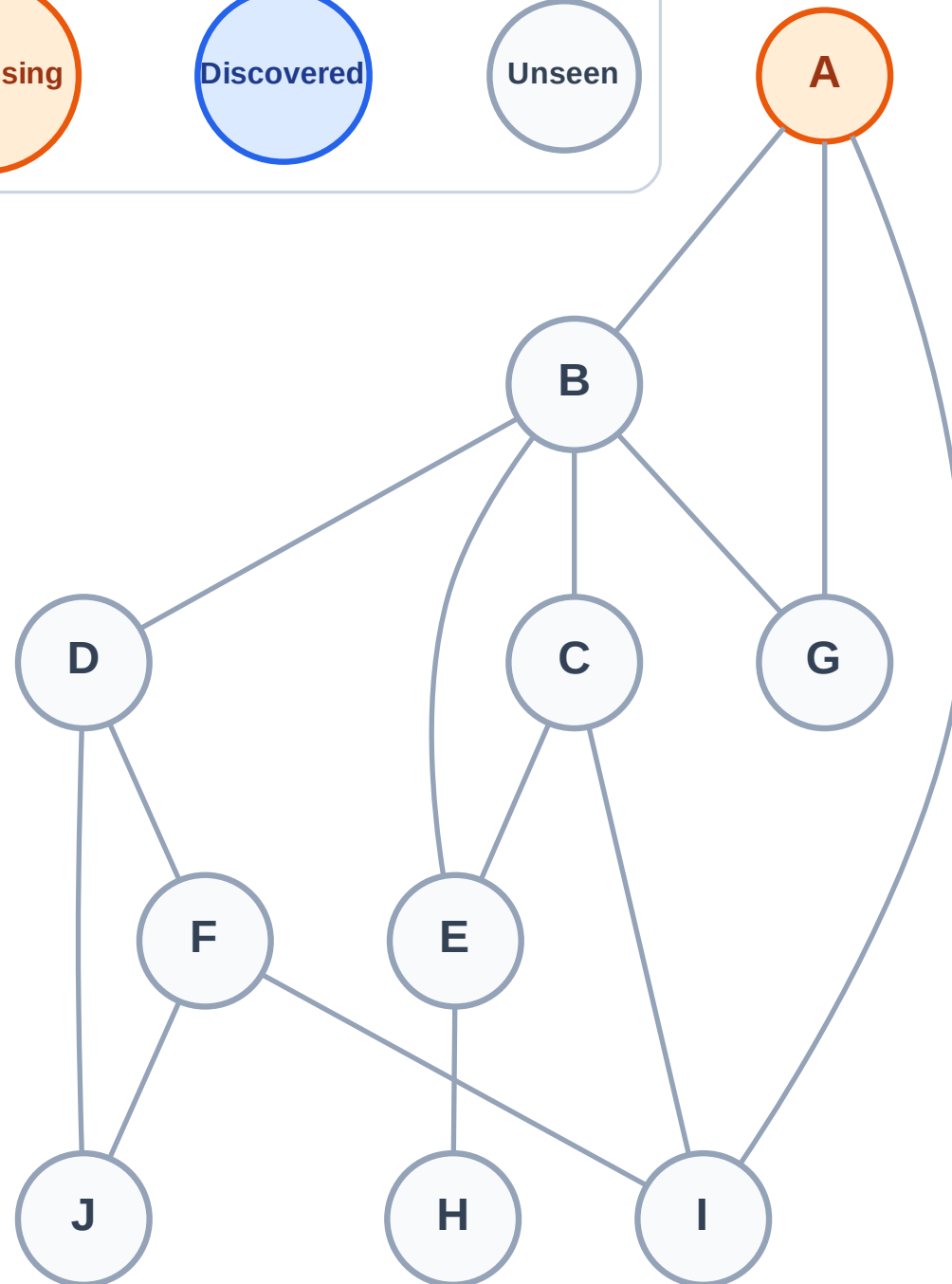
Step 2: Process node A

Legend



Queue: (empty)

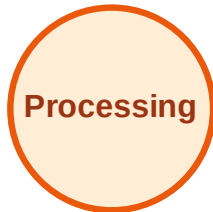
Visited: (none)



BFS Traversal

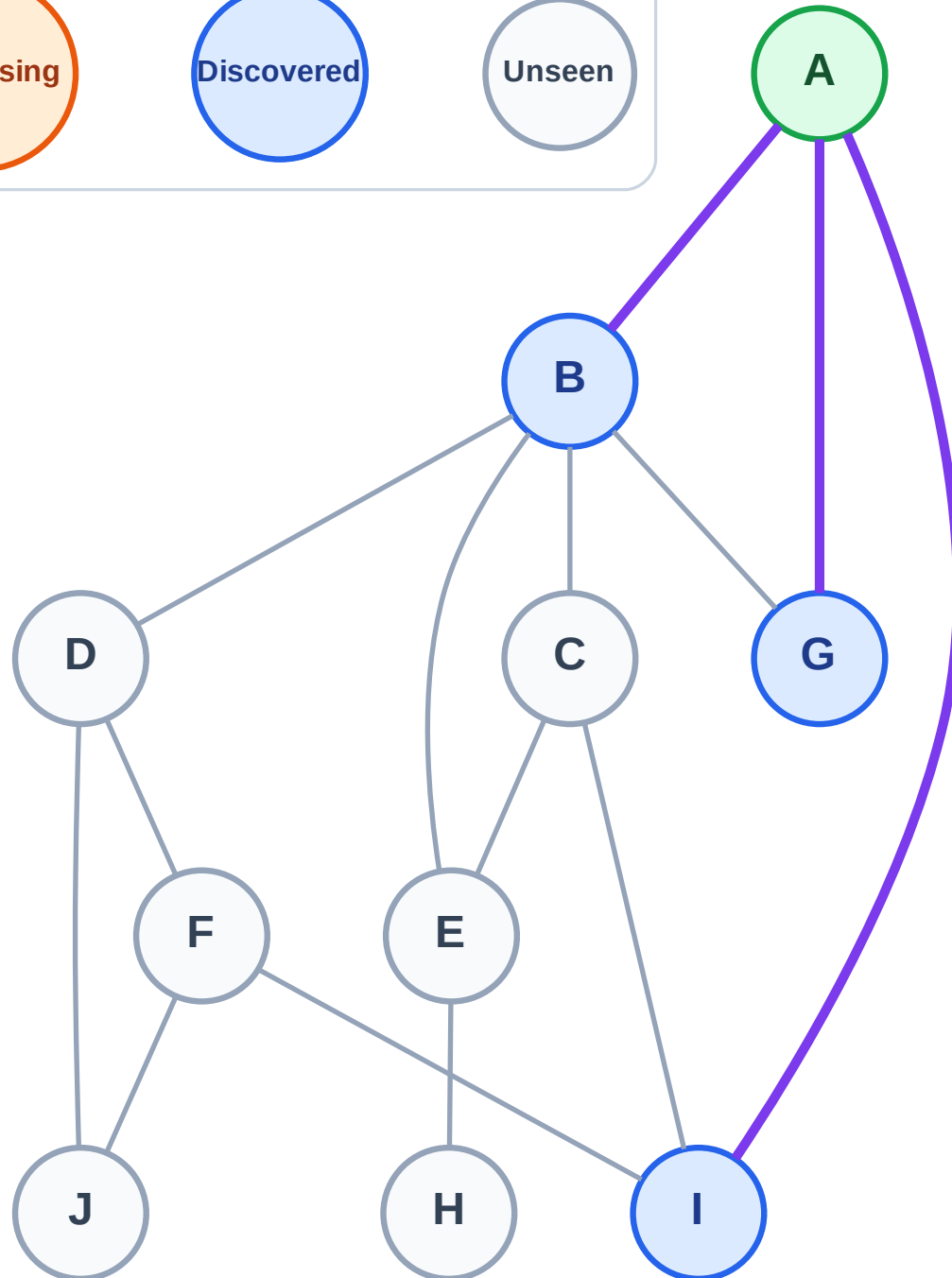
Step 3: Discover B, G, I from A

Legend



Queue: B → G → I

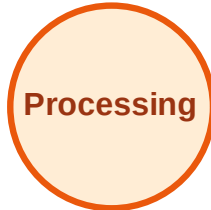
Visited: A



BFS Traversal

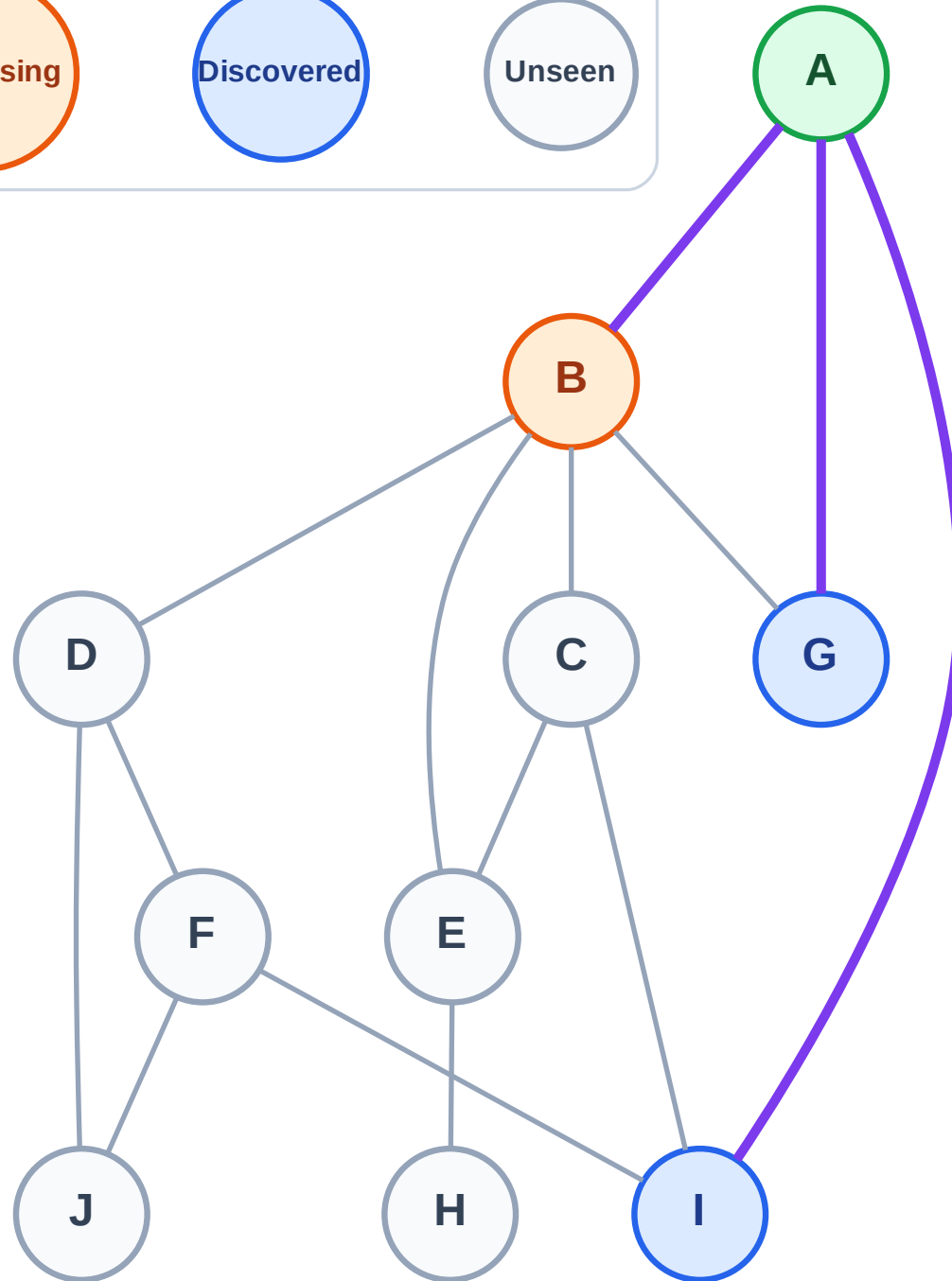
Step 4: Process node B

Legend



Queue: G → I

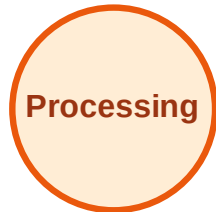
Visited: A



BFS Traversal

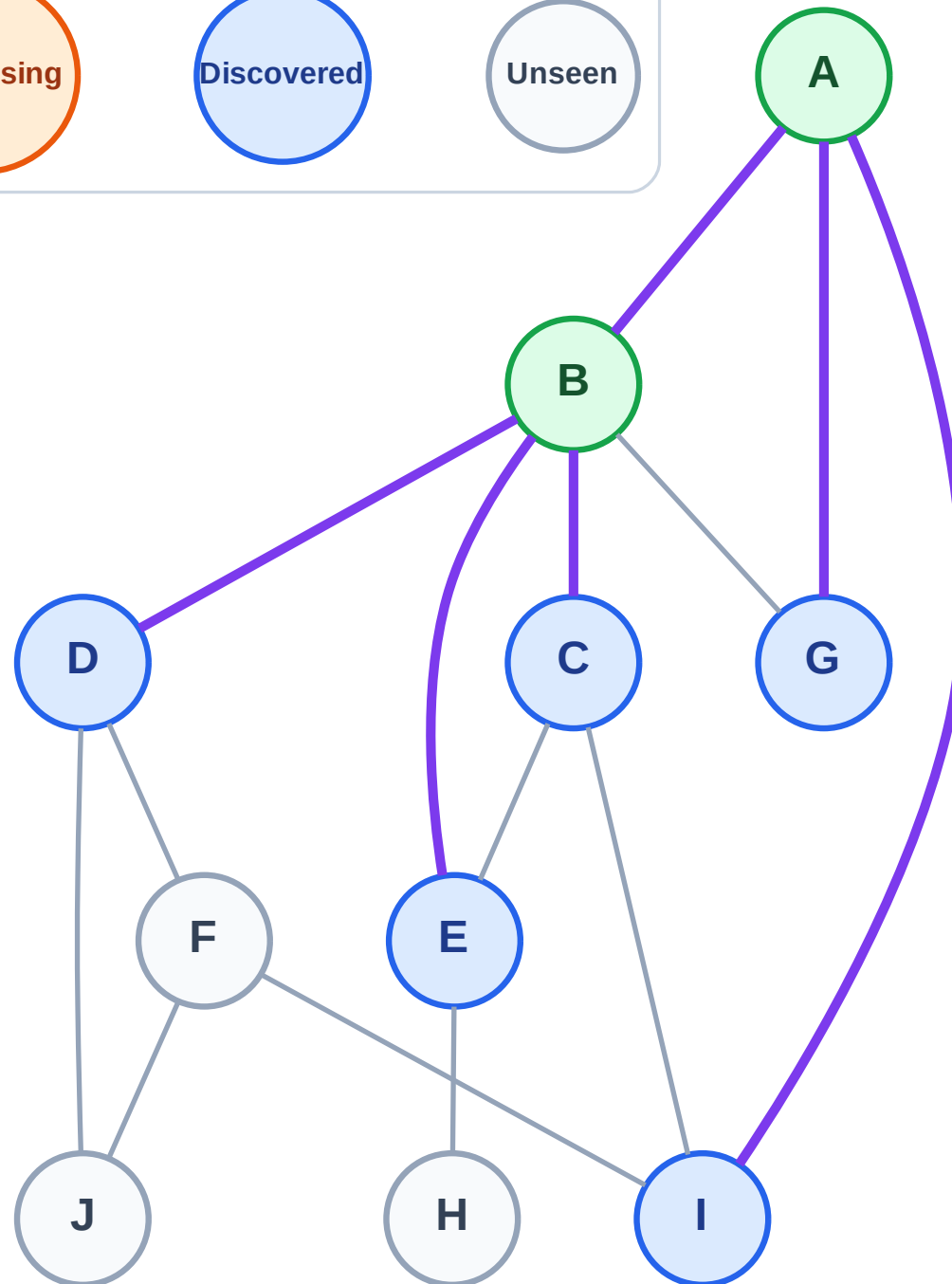
Step 5: Discover C, D, E from B

Legend



Queue: G → I → C → D → E

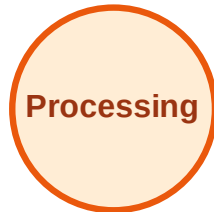
Visited: A, B



BFS Traversal

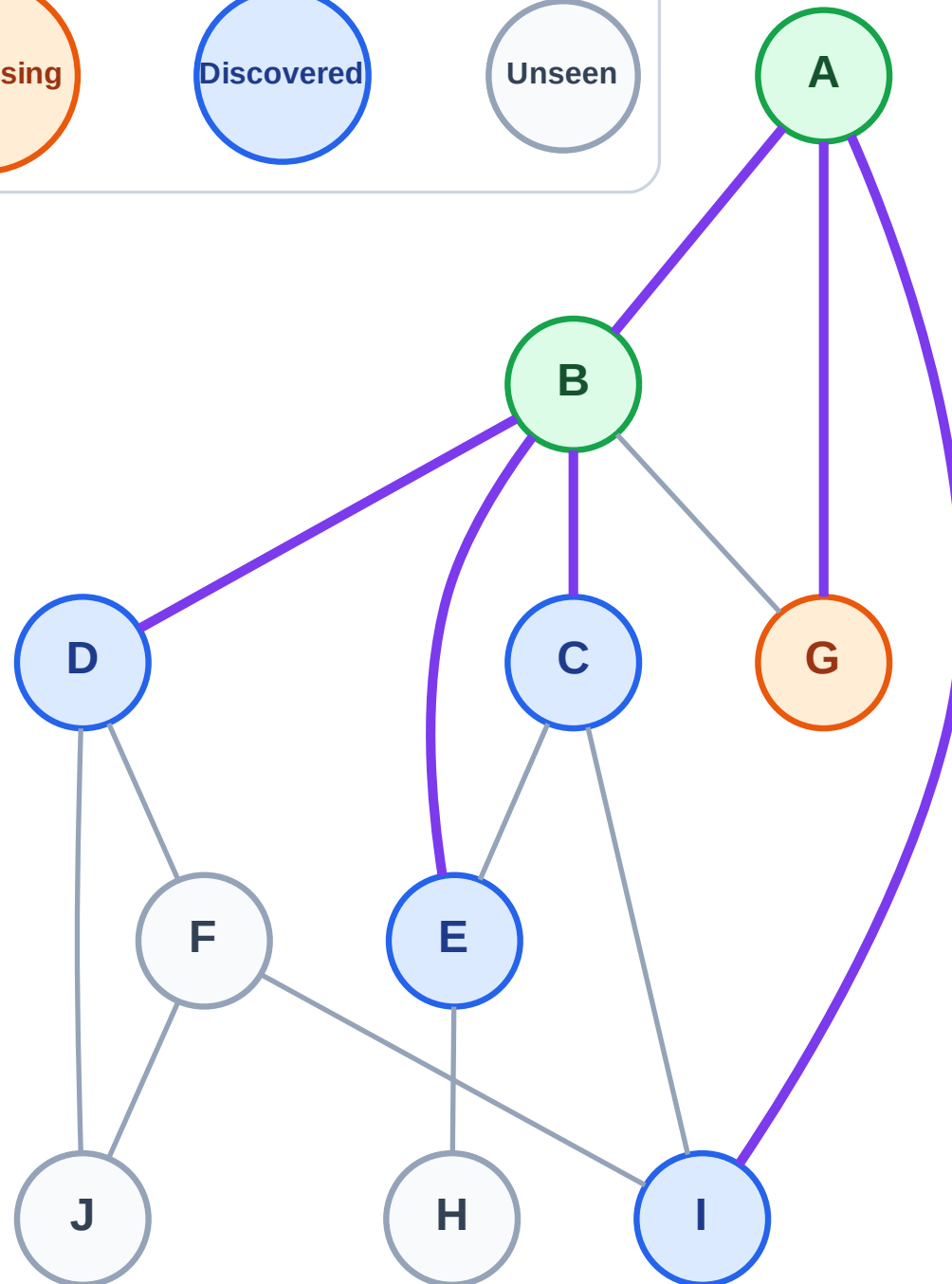
Step 6: Process node G

Legend



Queue: I → C → D → E

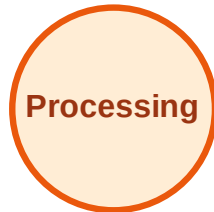
Visited: A, B



BFS Traversal

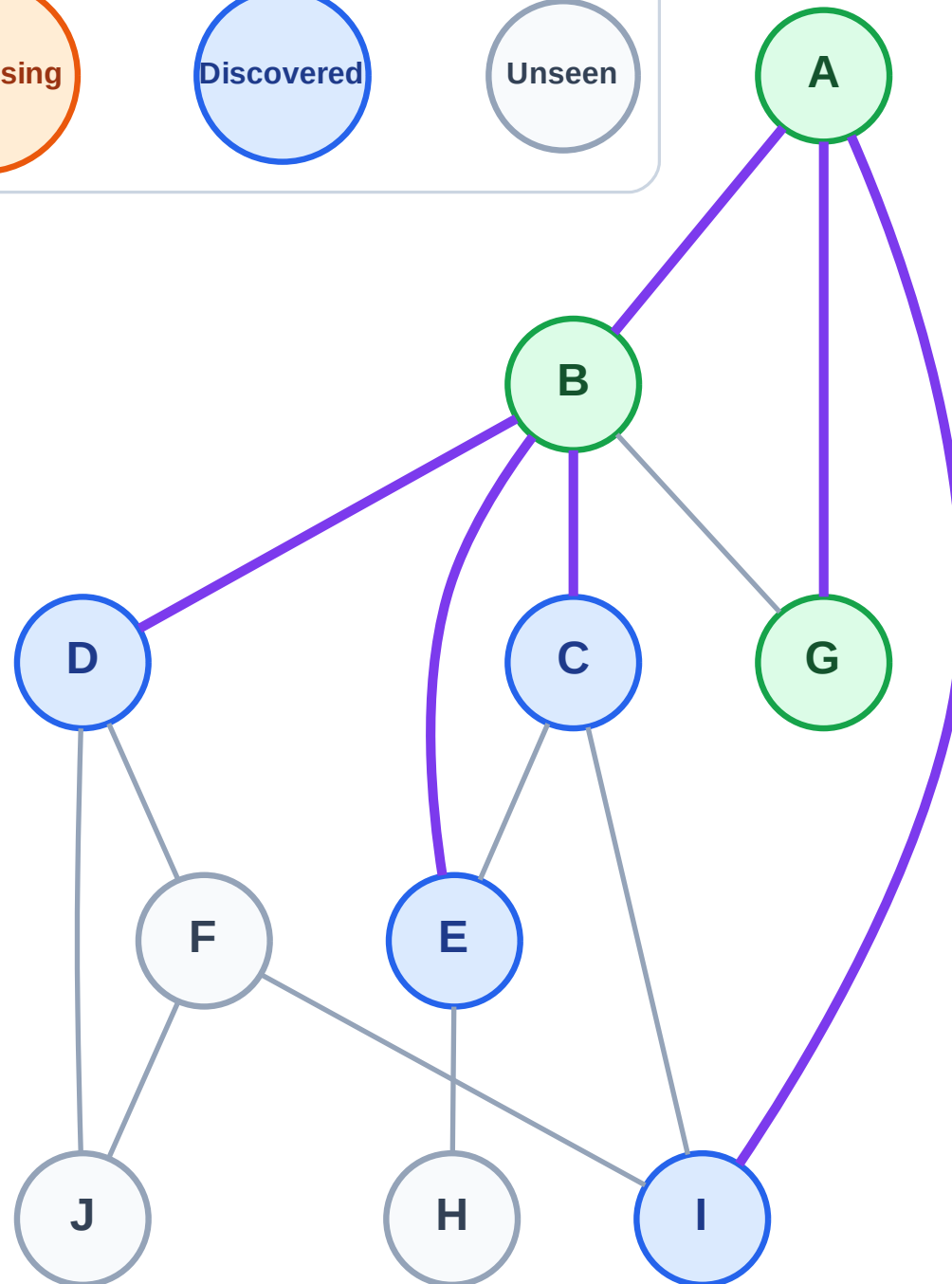
Step 7: No new neighbors from G

Legend



Queue: I → C → D → E

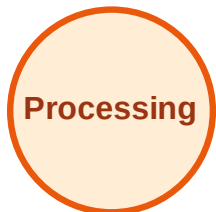
Visited: A, B, G



BFS Traversal

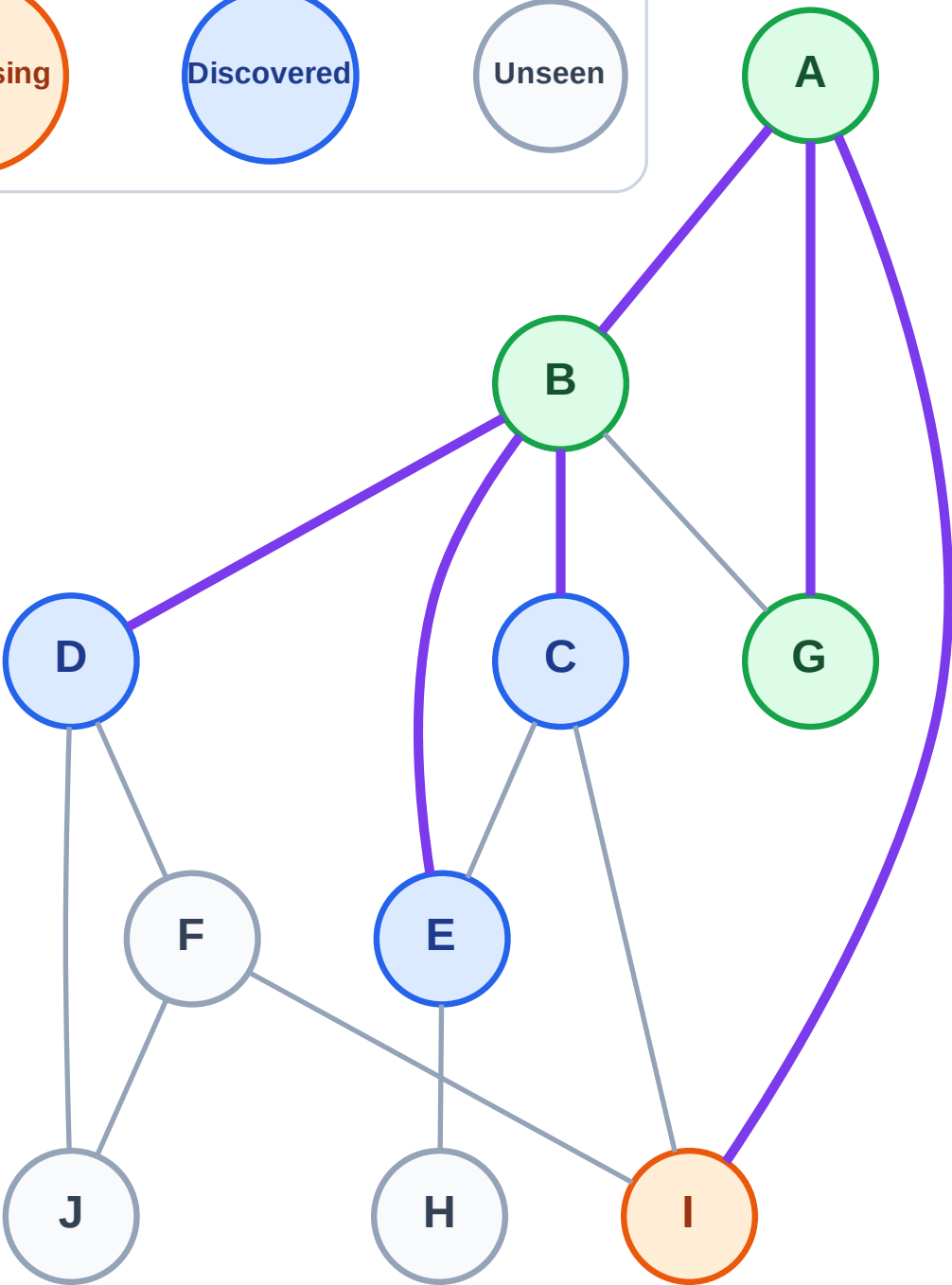
Step 8: Process node I

Legend



Queue: C → D → E

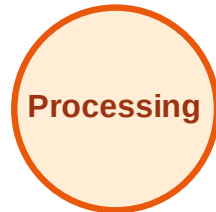
Visited: A, B, G



BFS Traversal

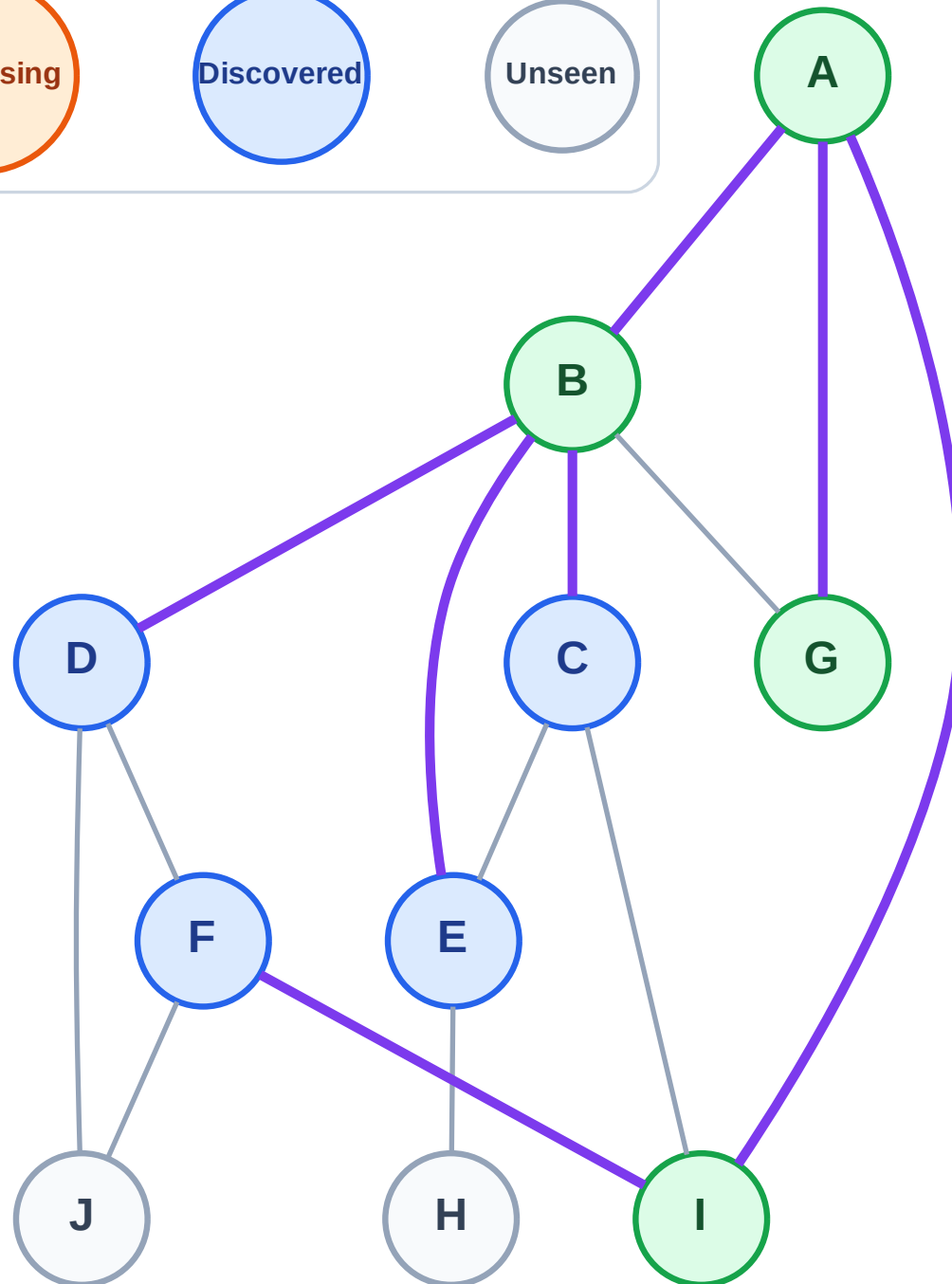
Step 9: Discover F from I

Legend



Queue: C → D → E → F

Visited: A, B, G, I



BFS Traversal

Step 10: Process node C

Legend

Complete

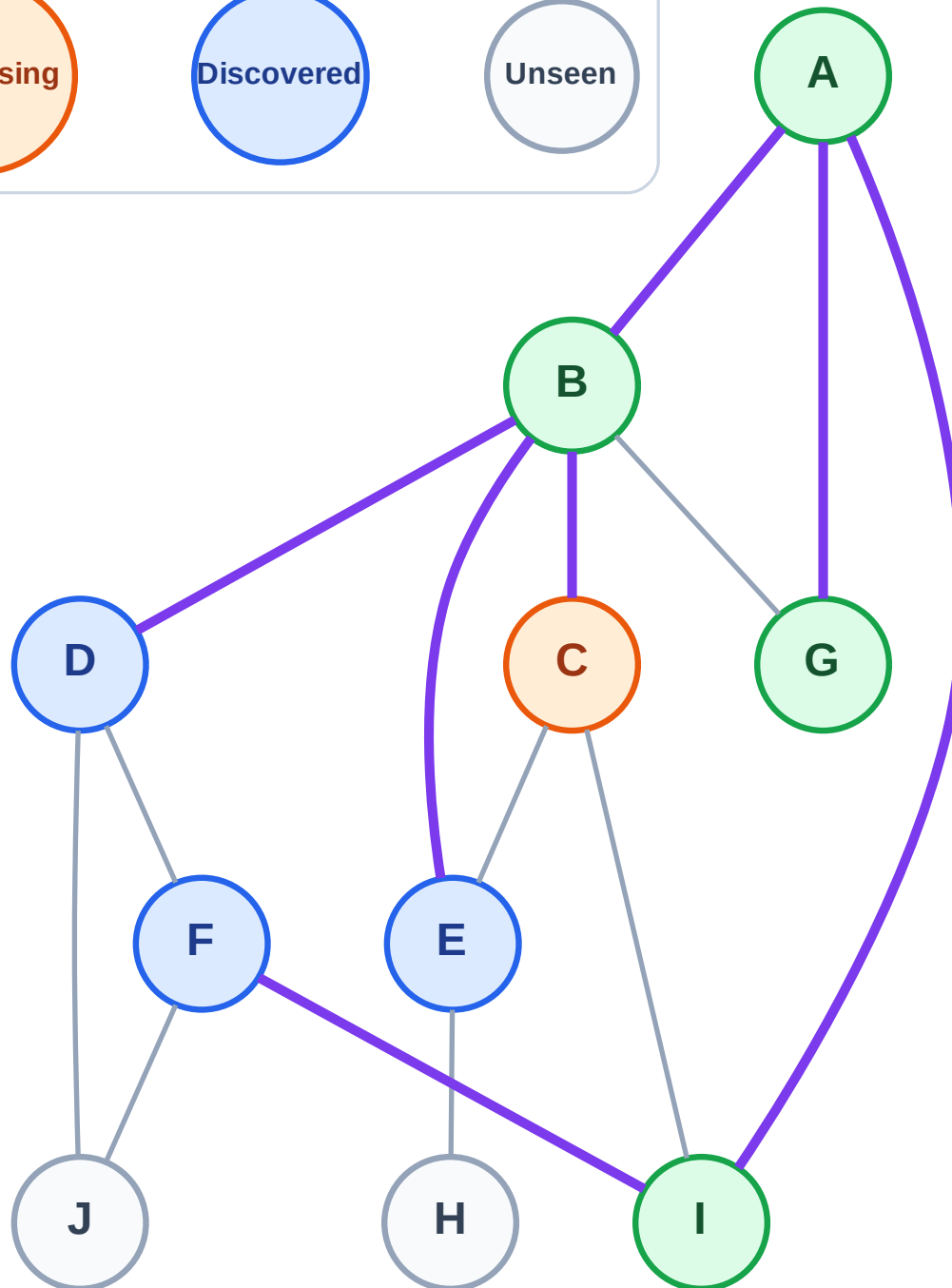
Processing

Discovered

Unseen

Queue: D → E → F

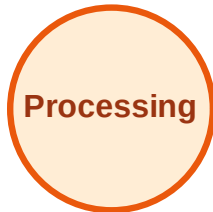
Visited: A, B, G, I



BFS Traversal

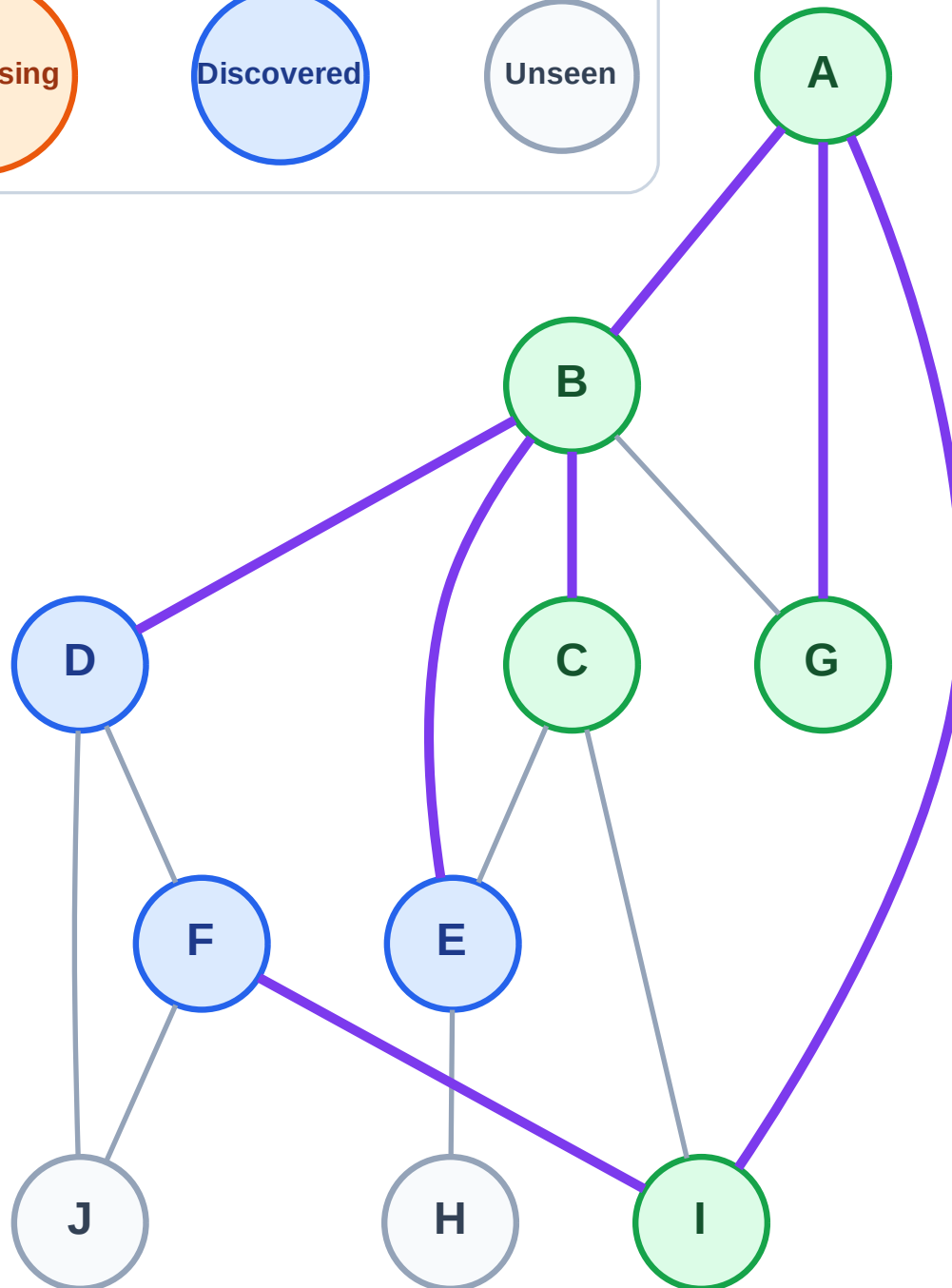
Step 11: No new neighbors from C

Legend



Queue: D → E → F

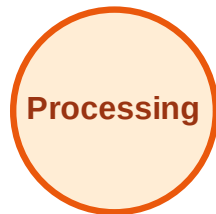
Visited: A, B, C, G, I



BFS Traversal

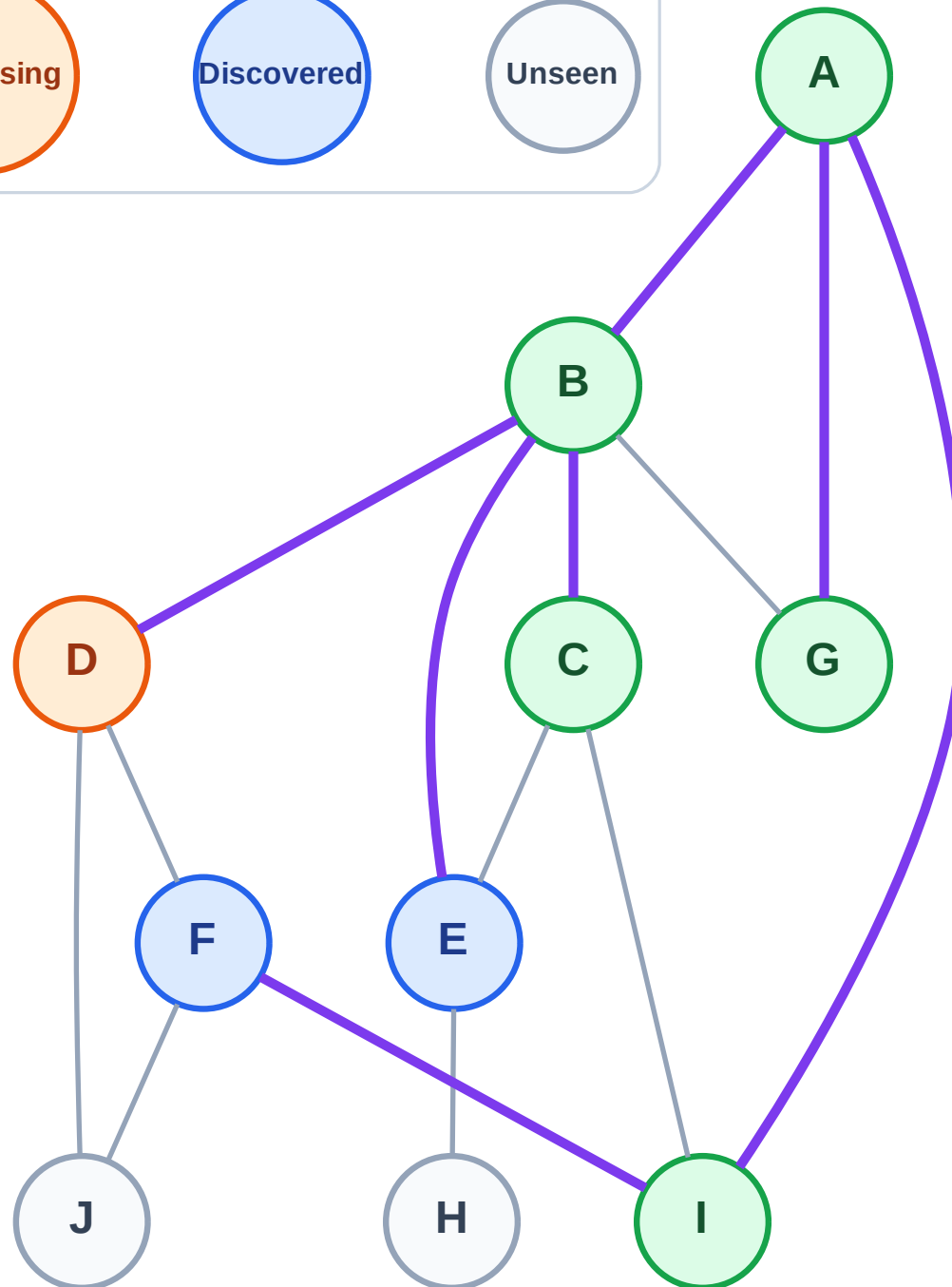
Step 12: Process node D

Legend



Queue: E → F

Visited: A, B, C, G, I



BFS Traversal

Step 13: Discover J from D

Legend

Complete

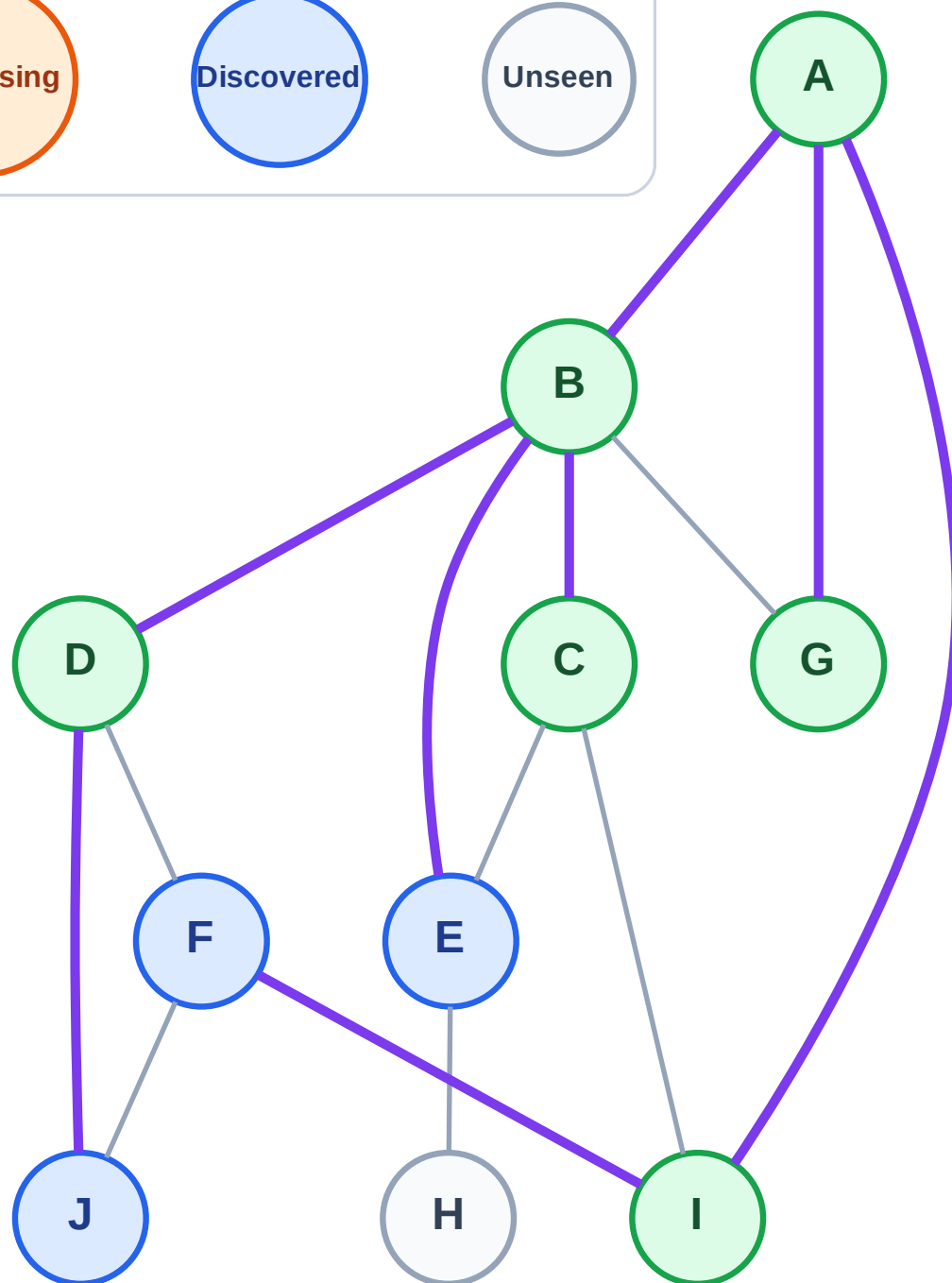
Processing

Discovered

Unseen

Queue: E → F → J

Visited: A, B, C, D, G, I



BFS Traversal

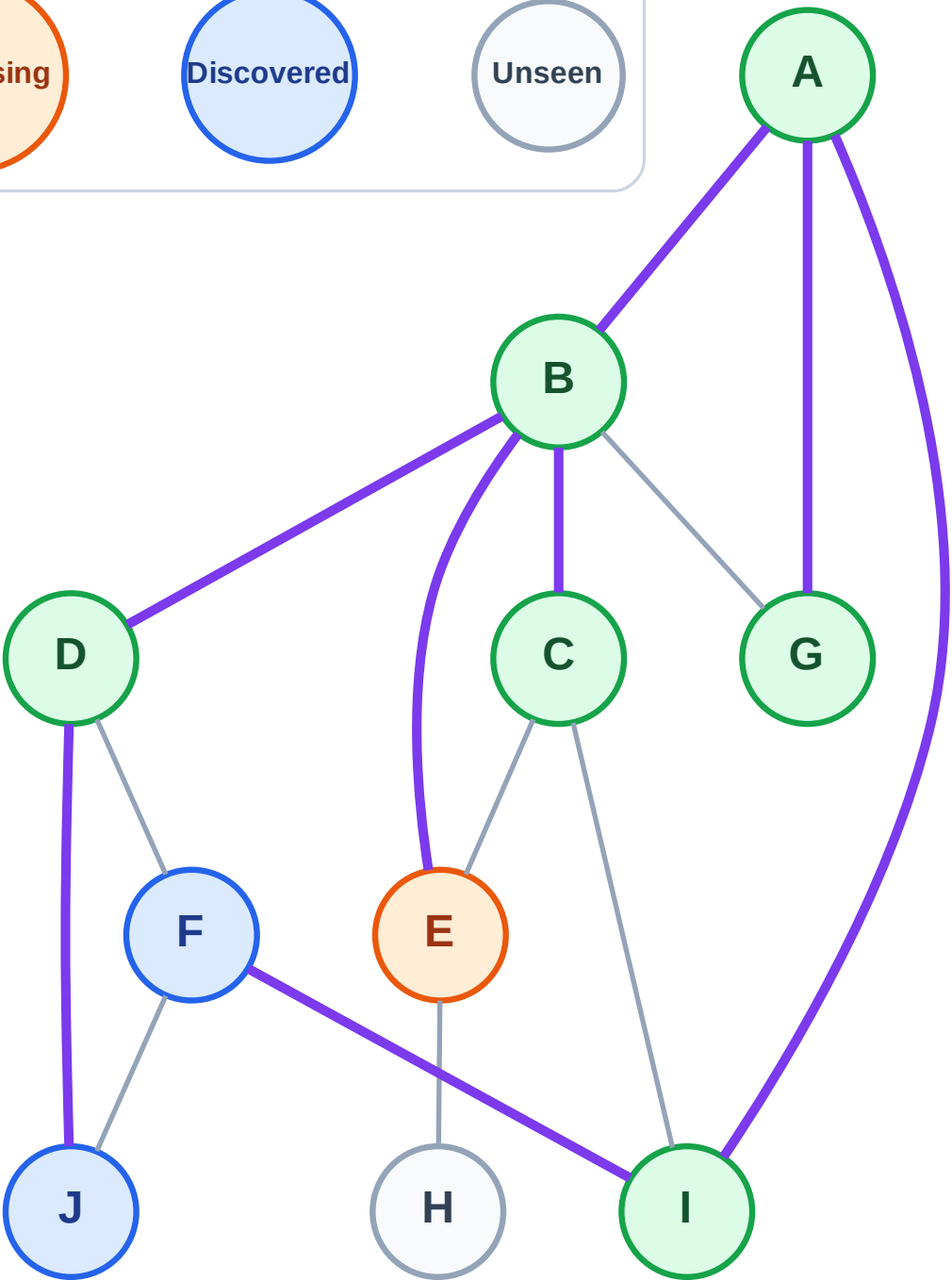
Step 14: Process node E

Legend



Queue: F → J

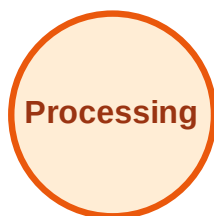
Visited: A, B, C, D, G, I



BFS Traversal

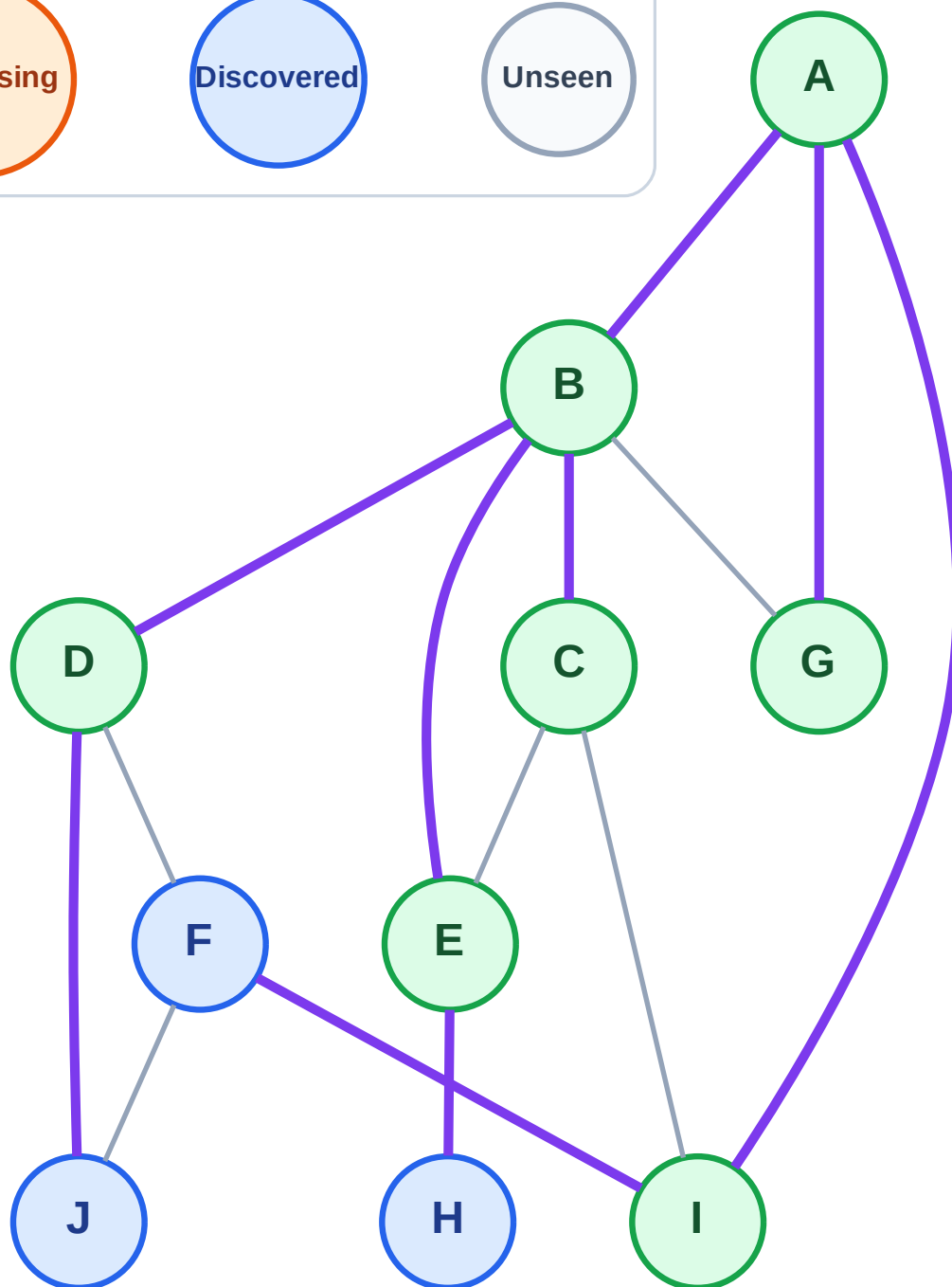
Step 15: Discover H from E

Legend



Queue: F → J → H

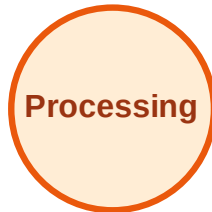
Visited: A, B, C, D, E, G, I



BFS Traversal

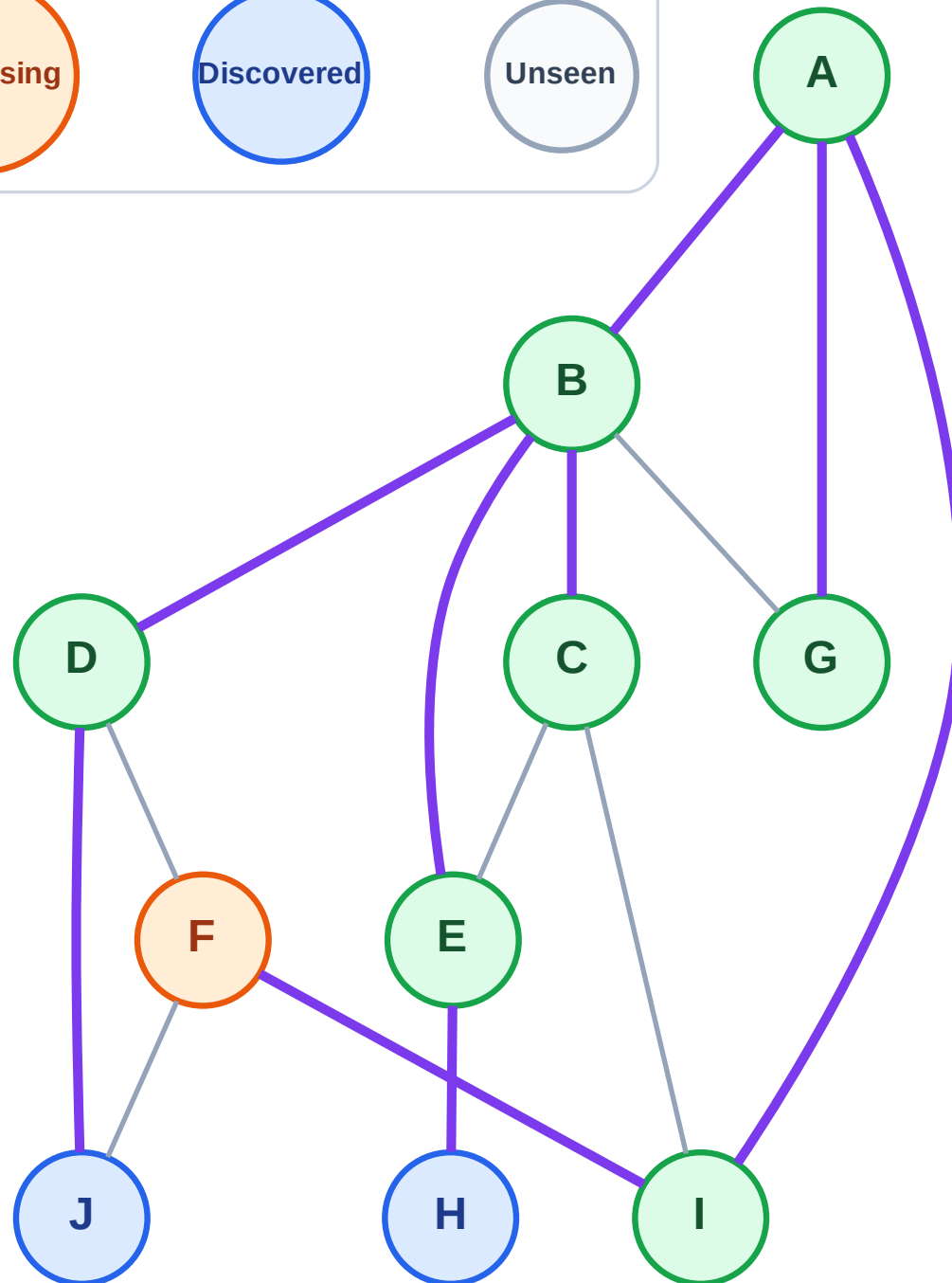
Step 16: Process node F

Legend



Queue: J → H

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 17: No new neighbors from F

Legend

Complete

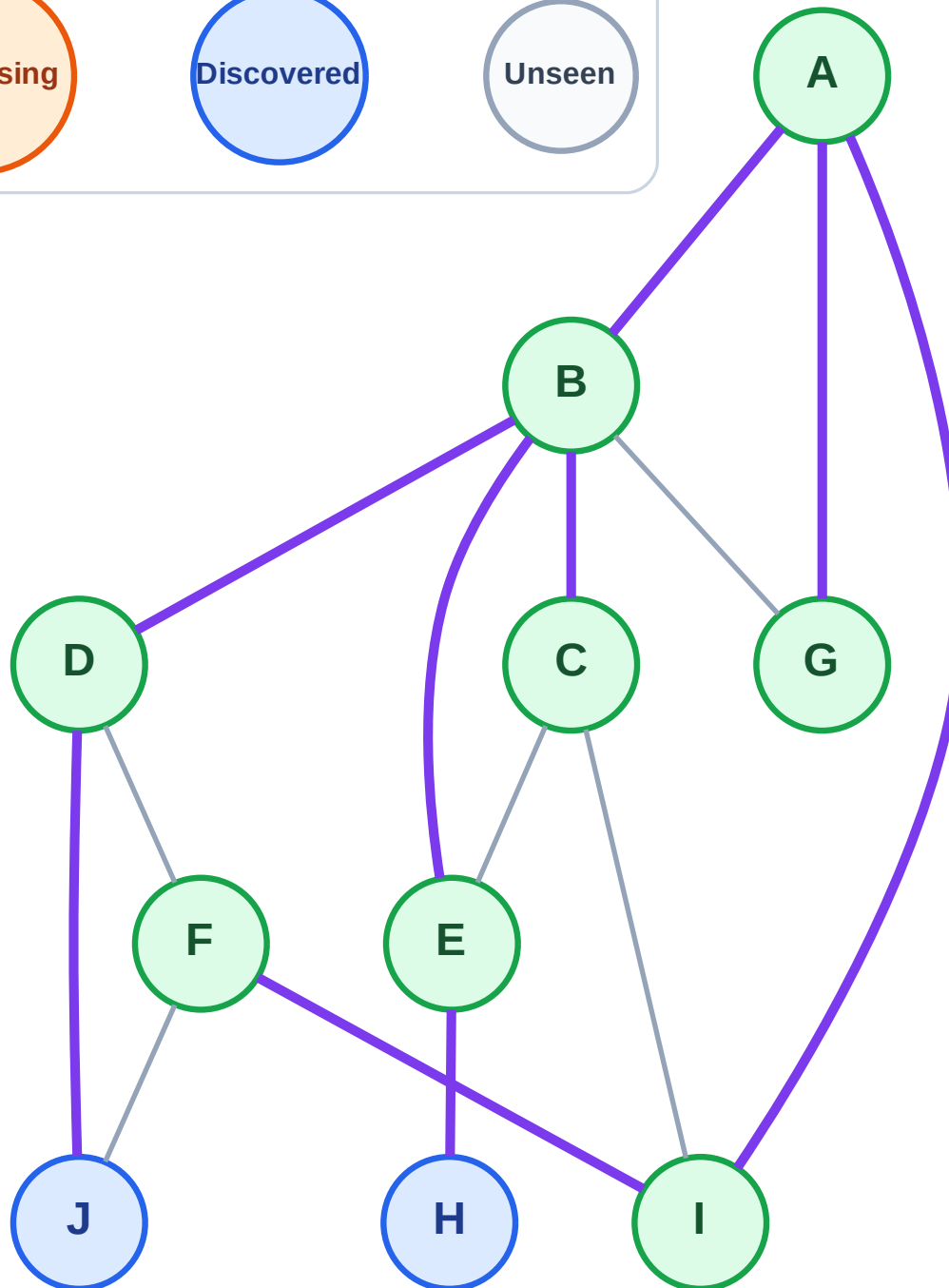
Processing

Discovered

Unseen

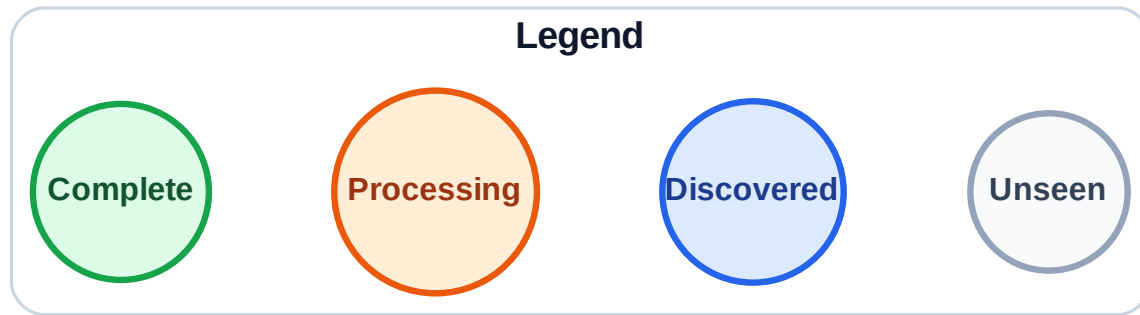
Queue: J → H

Visited: A, B, C, D, E, F, G, I



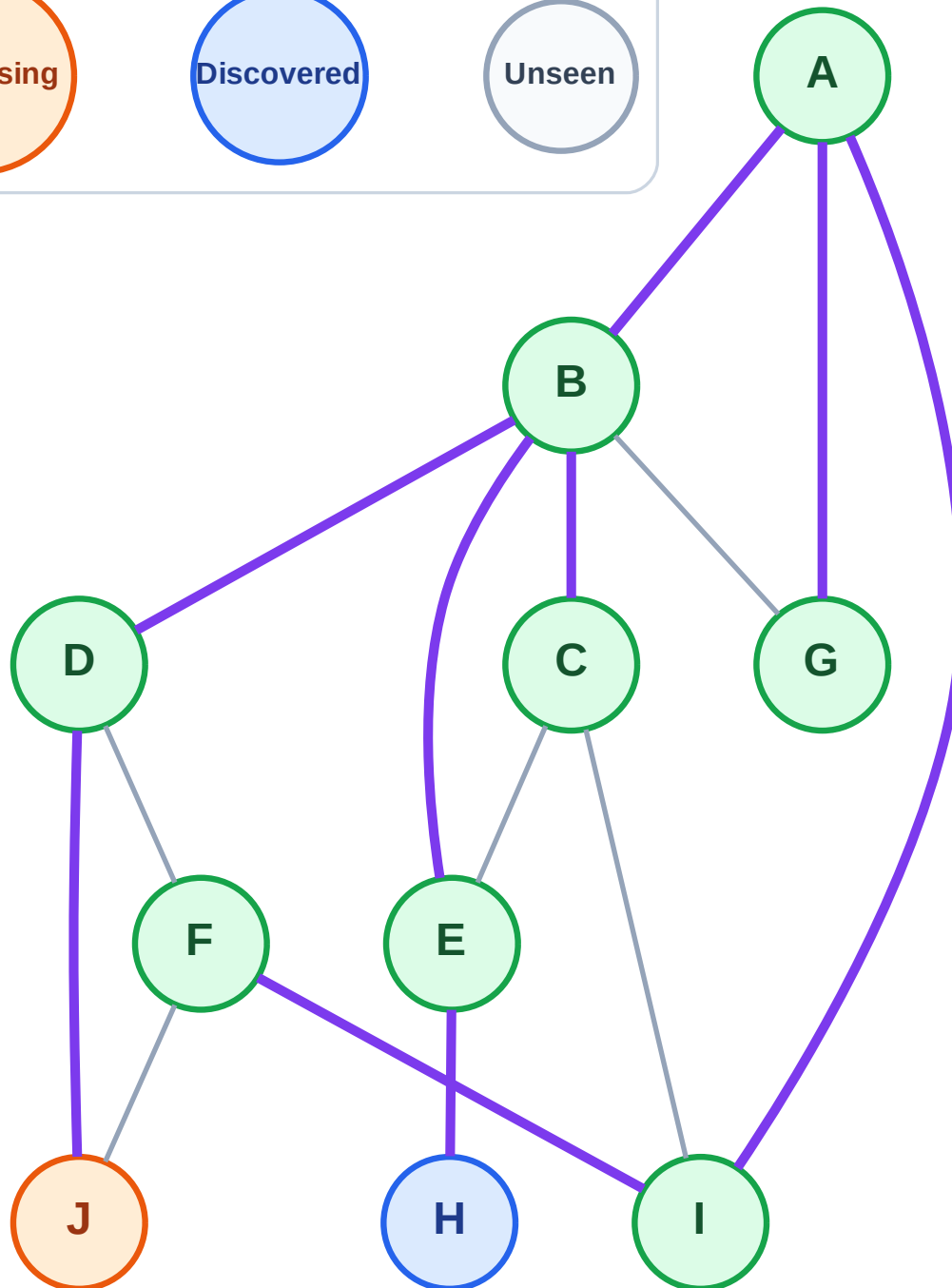
Step 18: Process node J

Step 18: Process node J



Queue: H

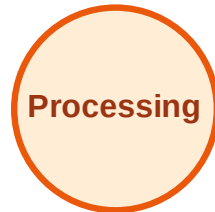
Visited: A, B, C, D, E, F, G, I



BFS Traversal

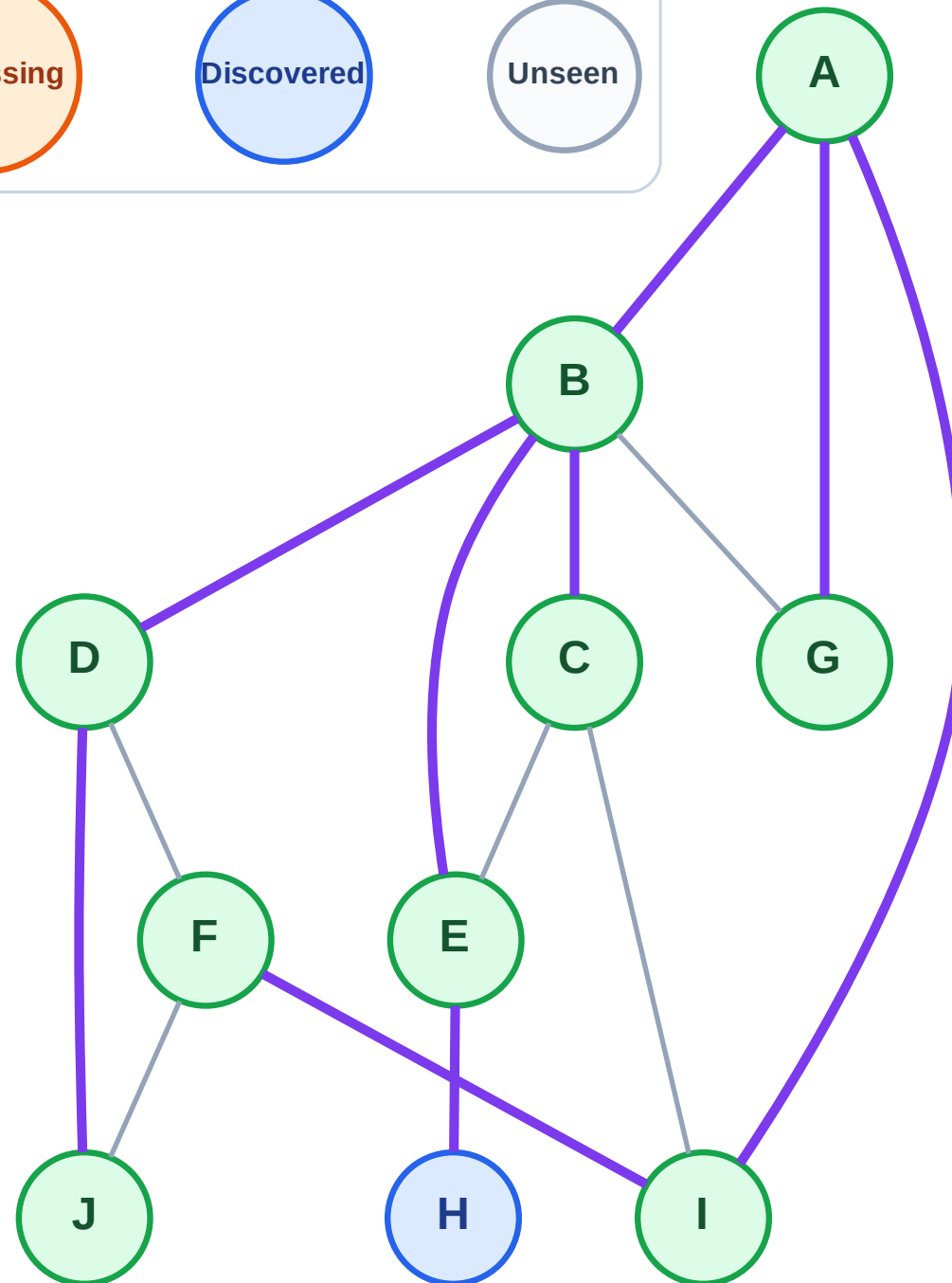
Step 19: No new neighbors from J

Legend



Queue: H

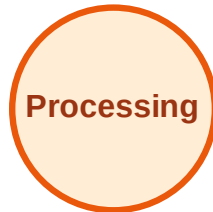
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

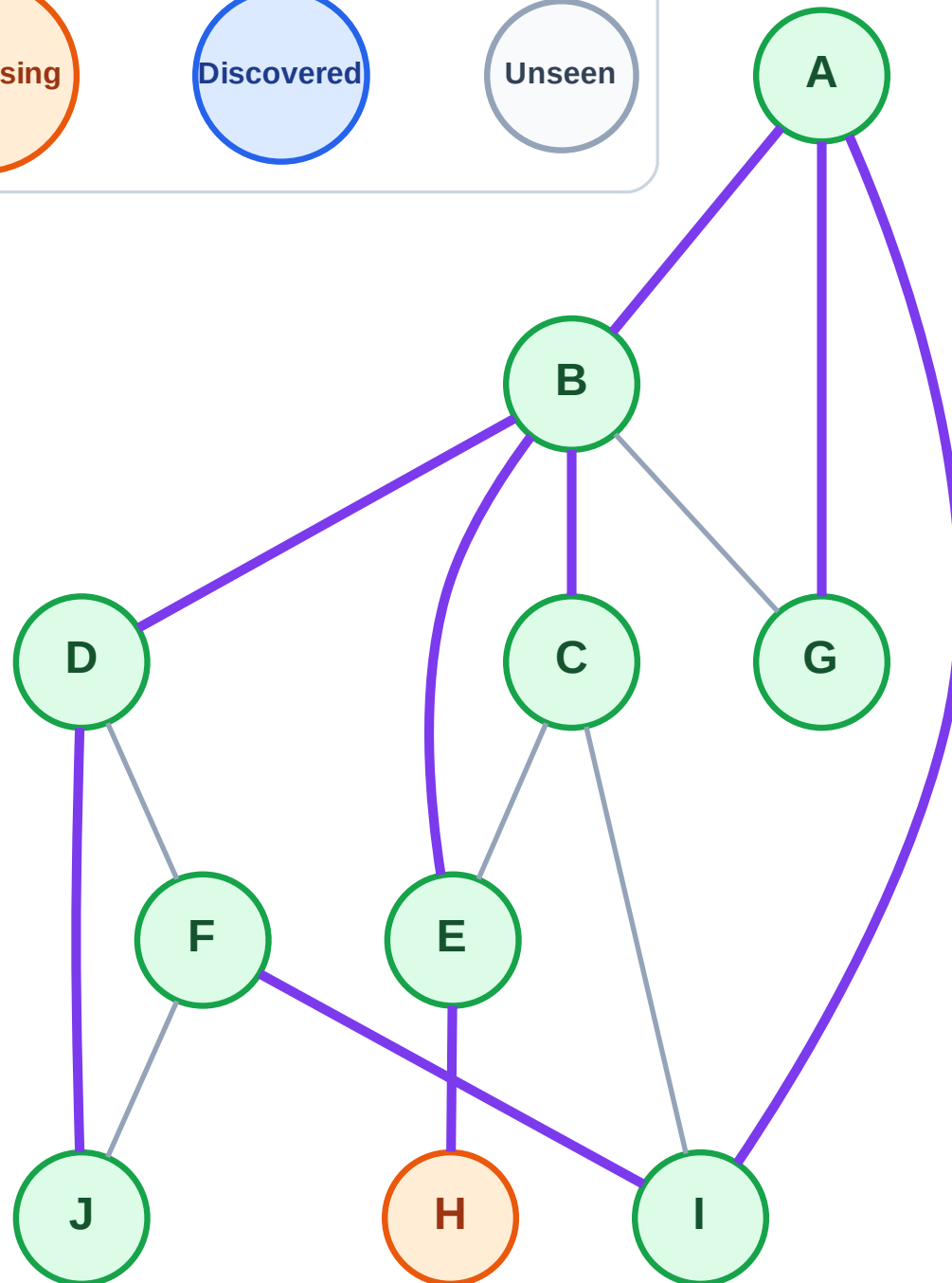
Step 20: Process node H

Legend



Queue: (empty)

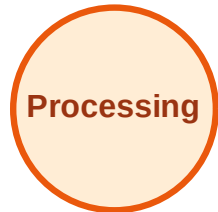
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

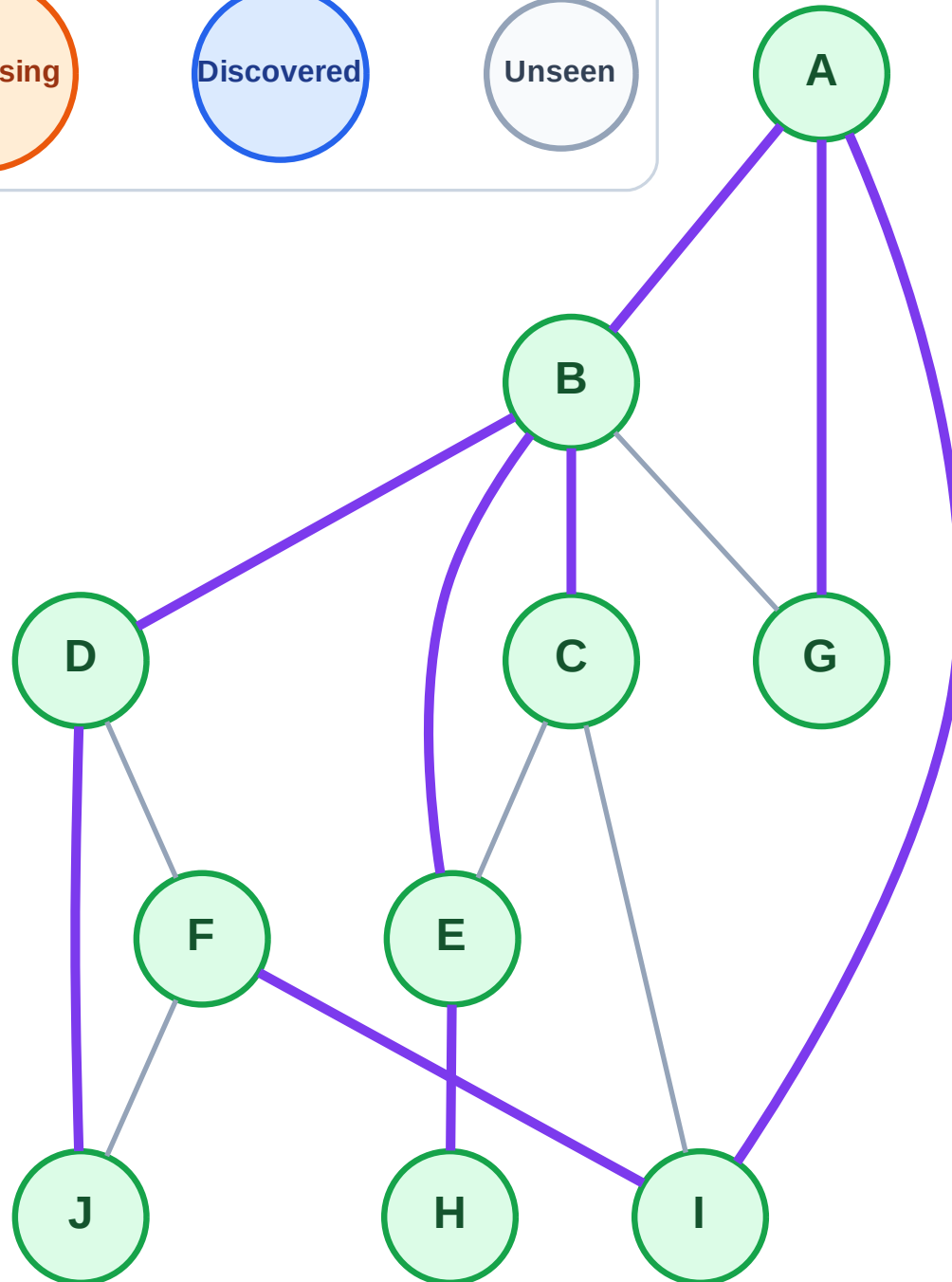
Step 21: No new neighbors from H

Legend



Queue: (empty)

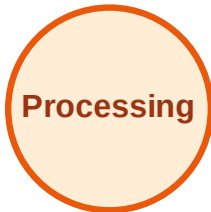
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

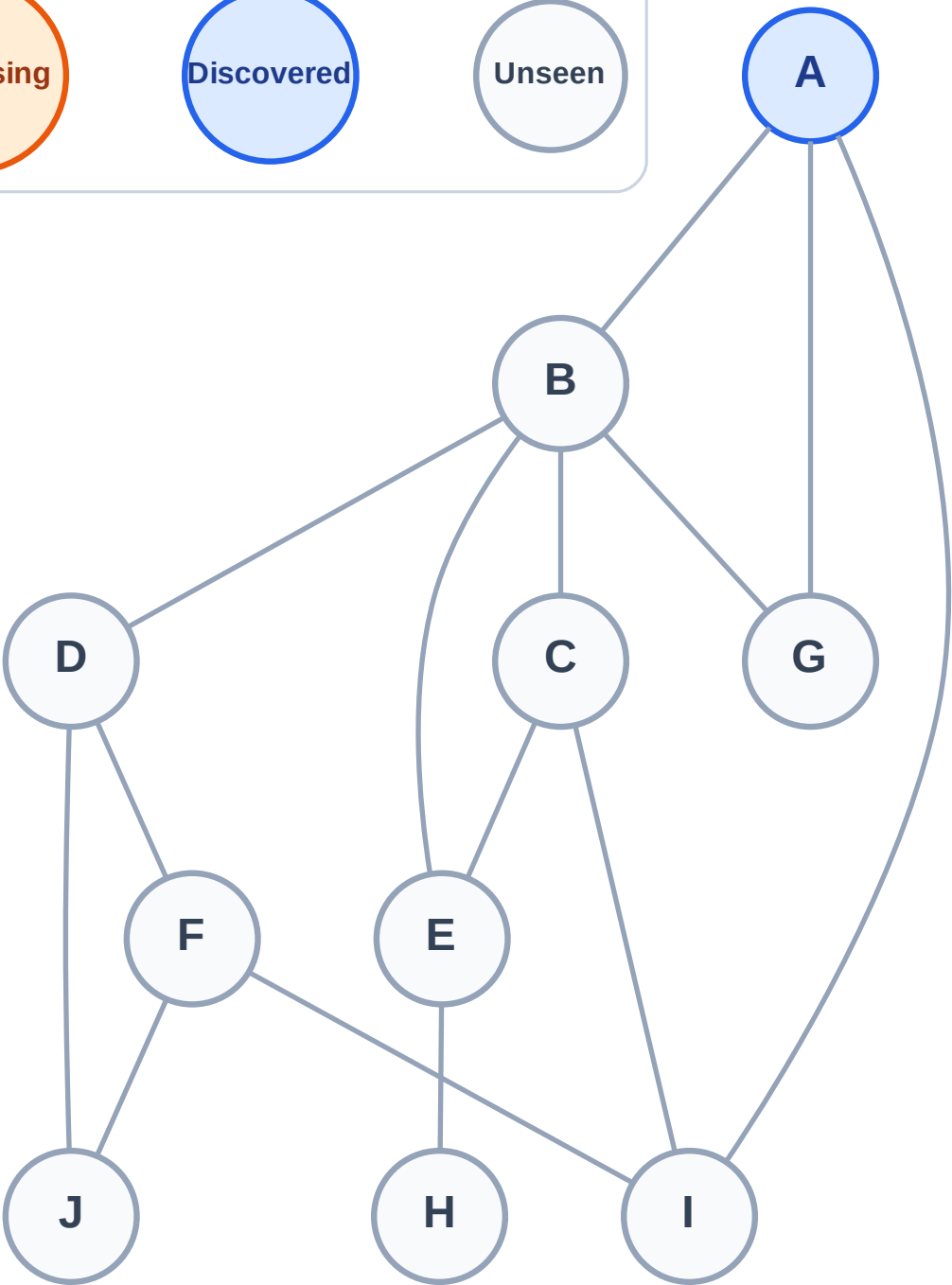
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

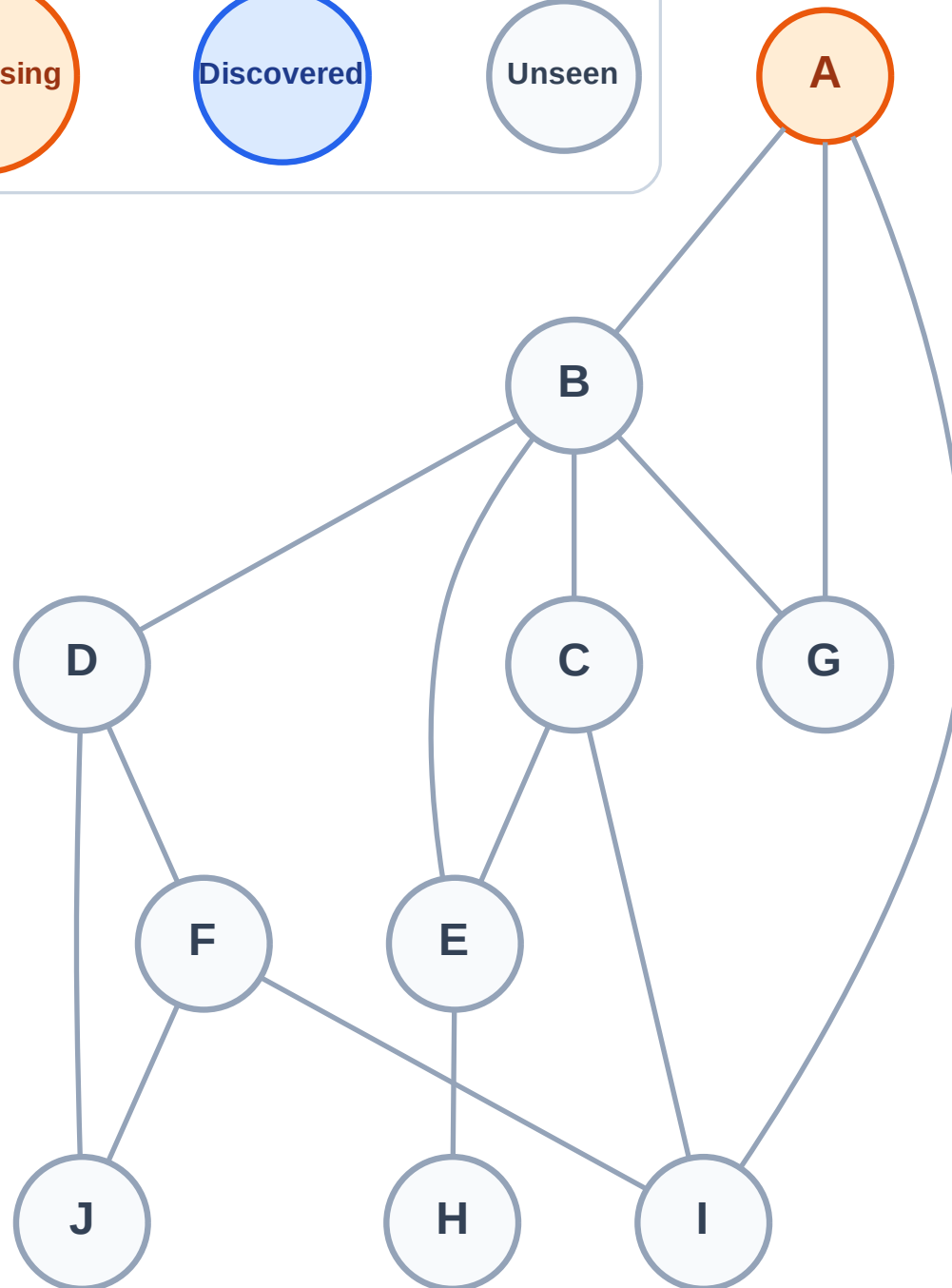
Processing

Discovered

Unseen

Stack (top at right): (empty)

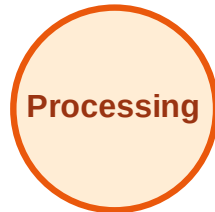
Visited: (none)



DFS Traversal

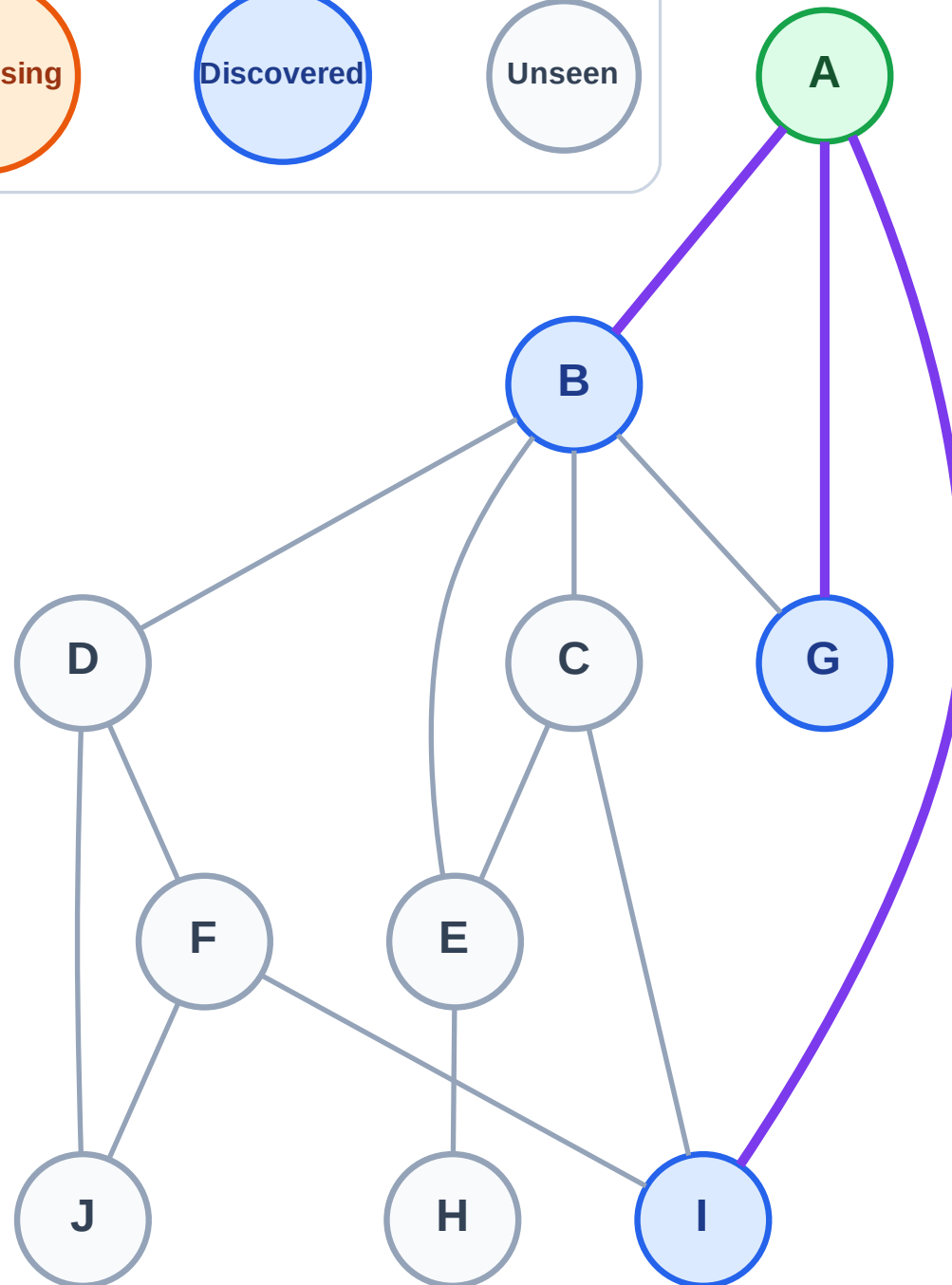
Step 3: Discover I, G, B from A

Legend



Stack (top at right): I | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

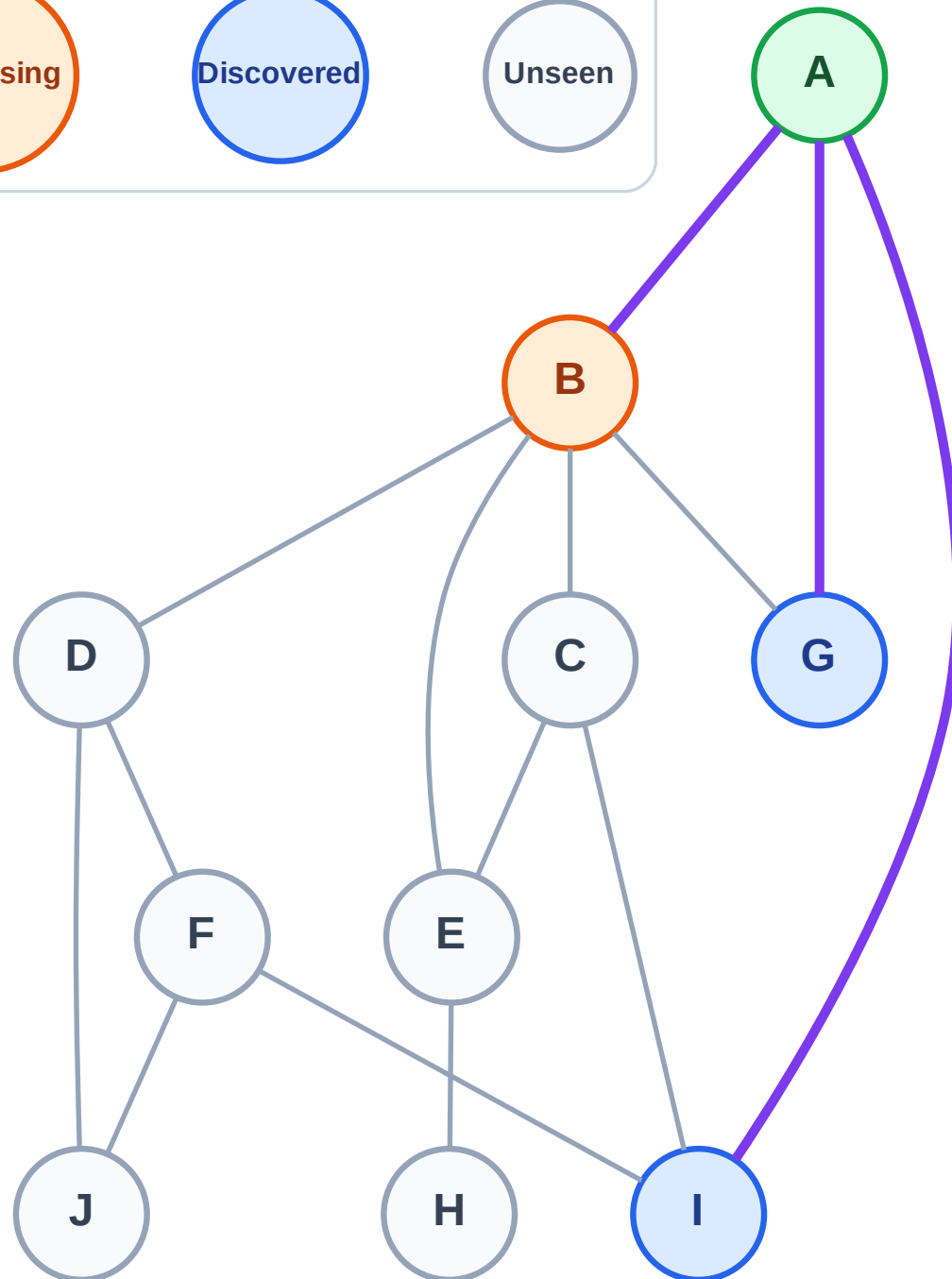
Processing

Discovered

Unseen

Stack (top at right): I | G

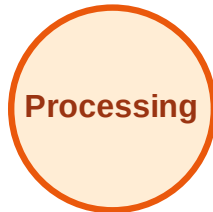
Visited: A



DFS Traversal

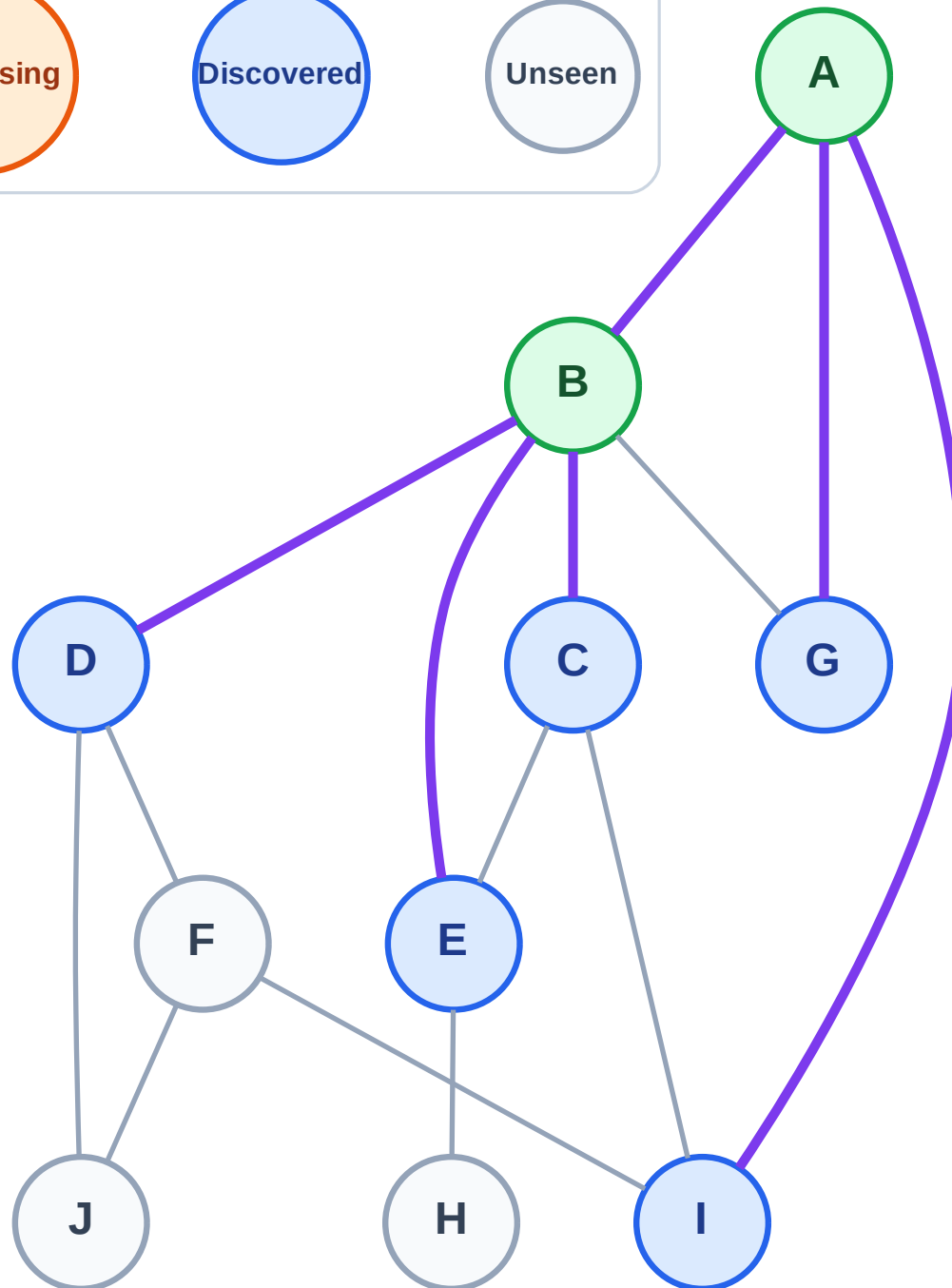
Step 5: Discover E, D, C from B

Legend



Stack (top at right): I | G | E | D | C

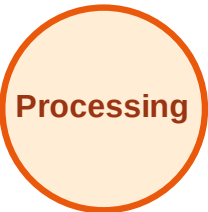
Visited: A, B



DFS Traversal

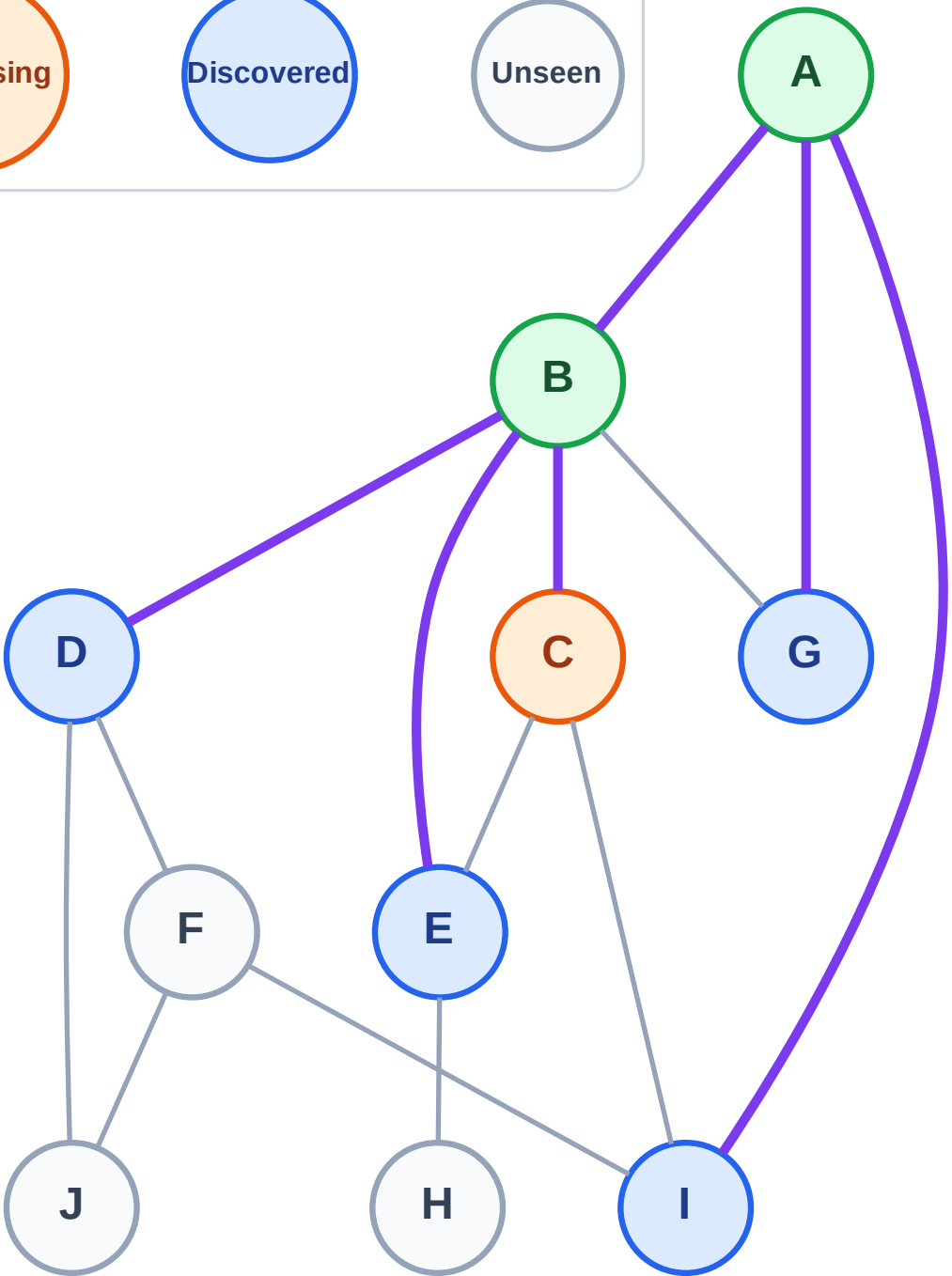
Step 6: Process node C

Legend



Stack (top at right): I | G | E | D

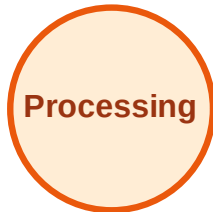
Visited: A, B



DFS Traversal

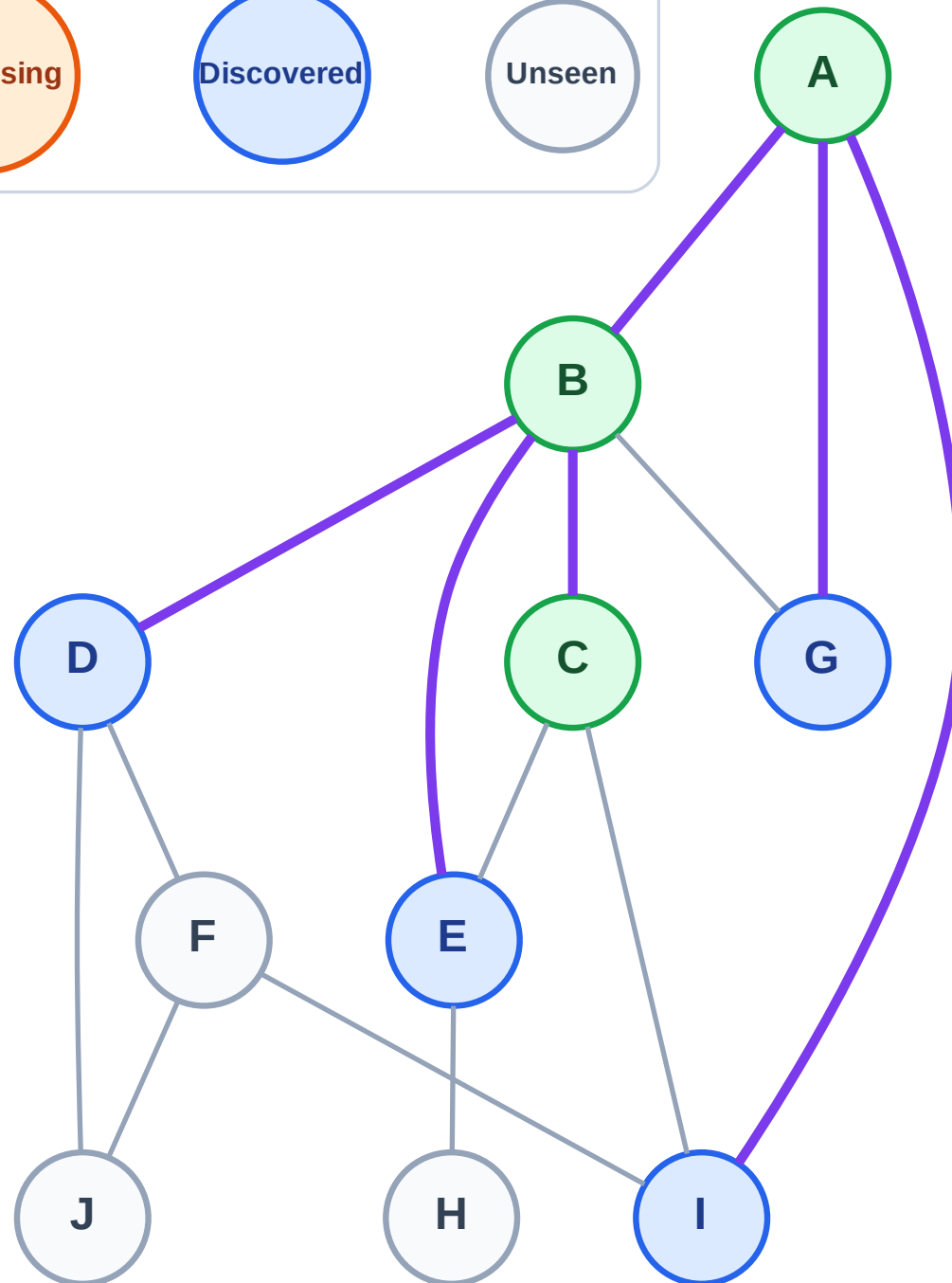
Step 7: No new neighbors from C

Legend



Stack (top at right): I | G | E | D

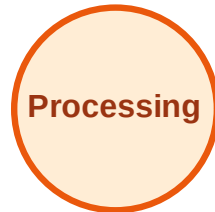
Visited: A, B, C



DFS Traversal

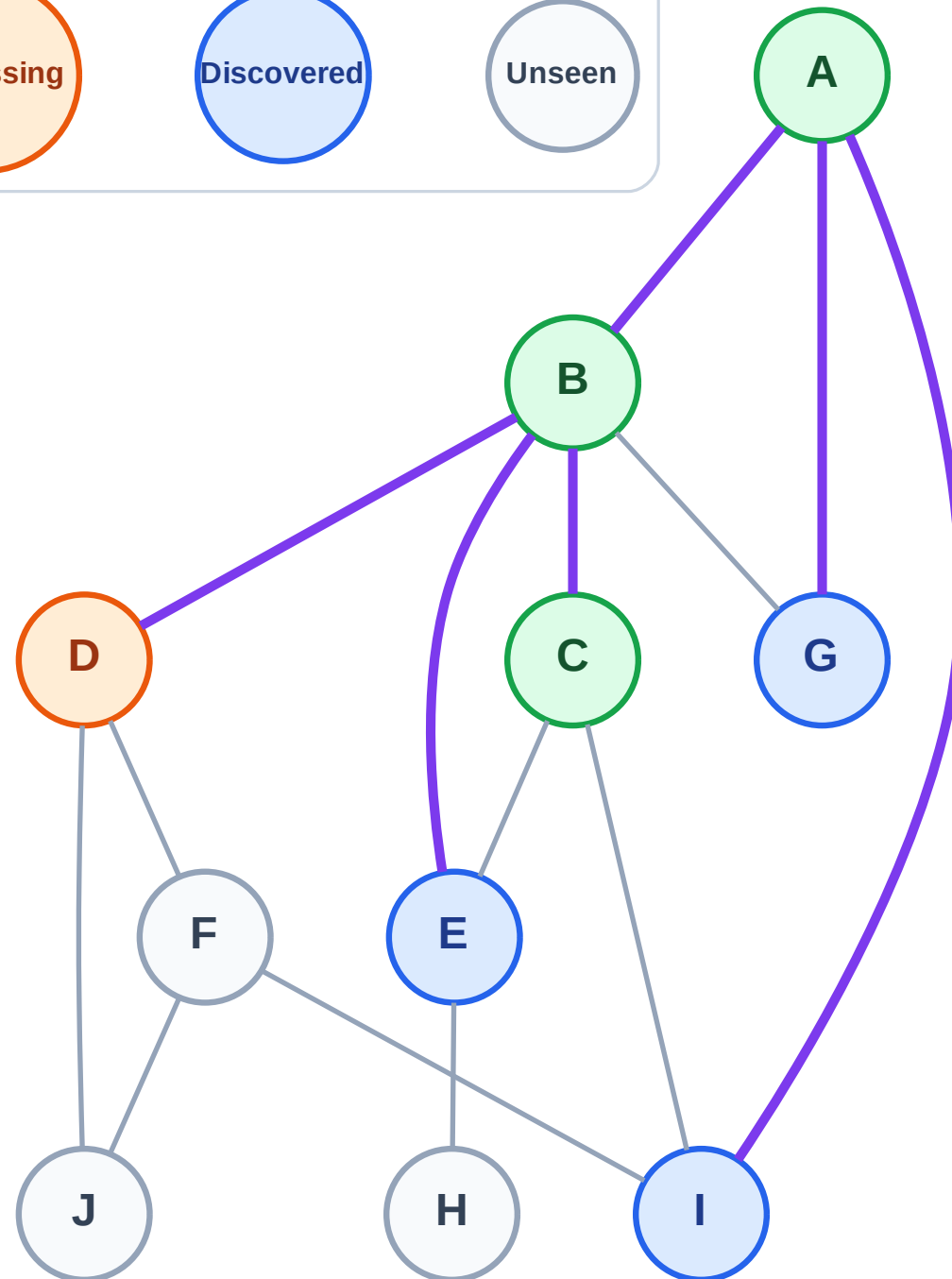
Step 8: Process node D

Legend



Stack (top at right): I | G | E

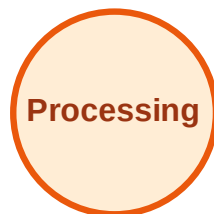
Visited: A, B, C



DFS Traversal

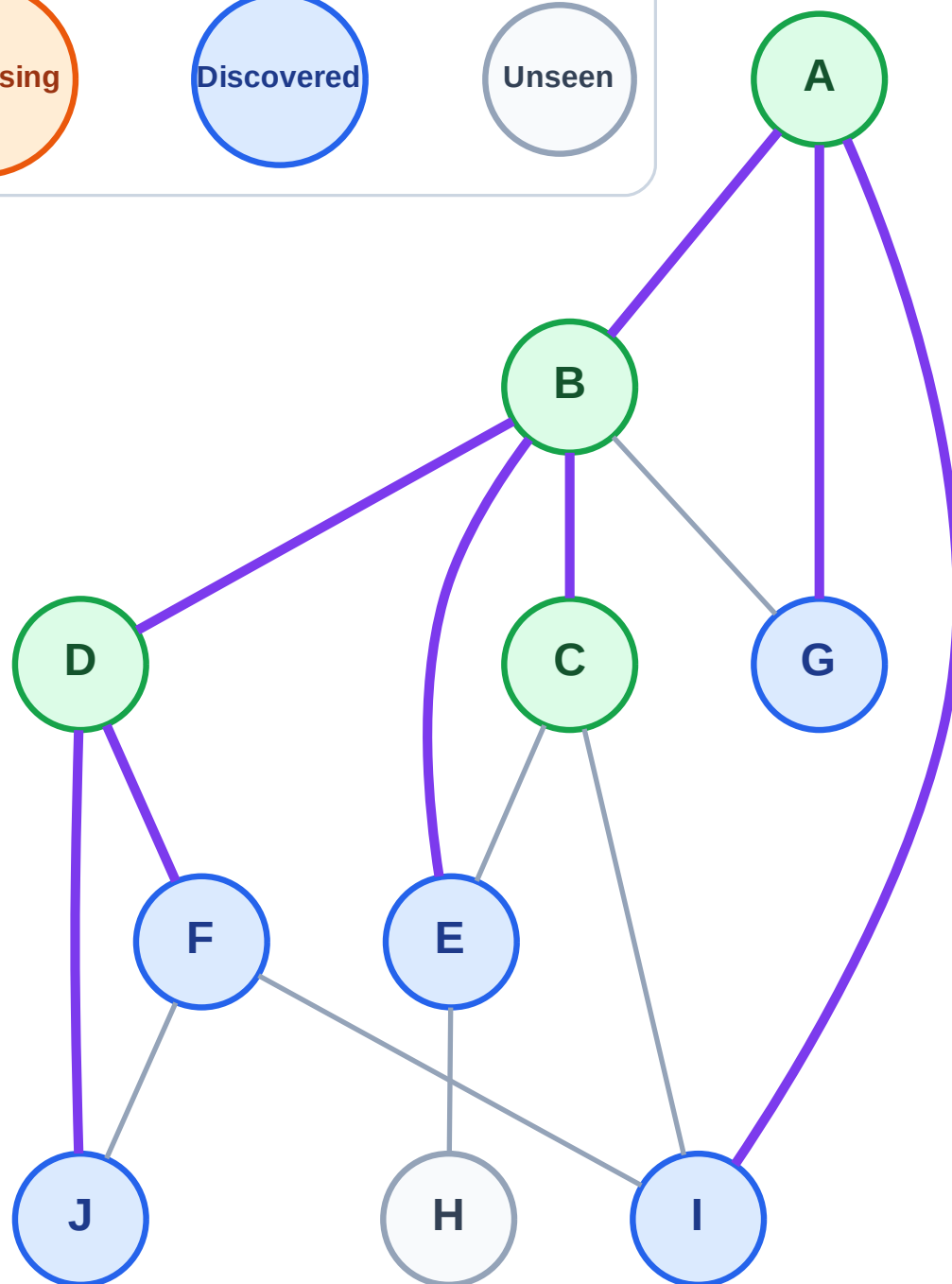
Step 9: Discover J, F from D

Legend



Stack (top at right): I | G | E | J | F

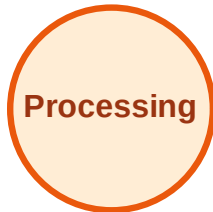
Visited: A, B, C, D



DFS Traversal

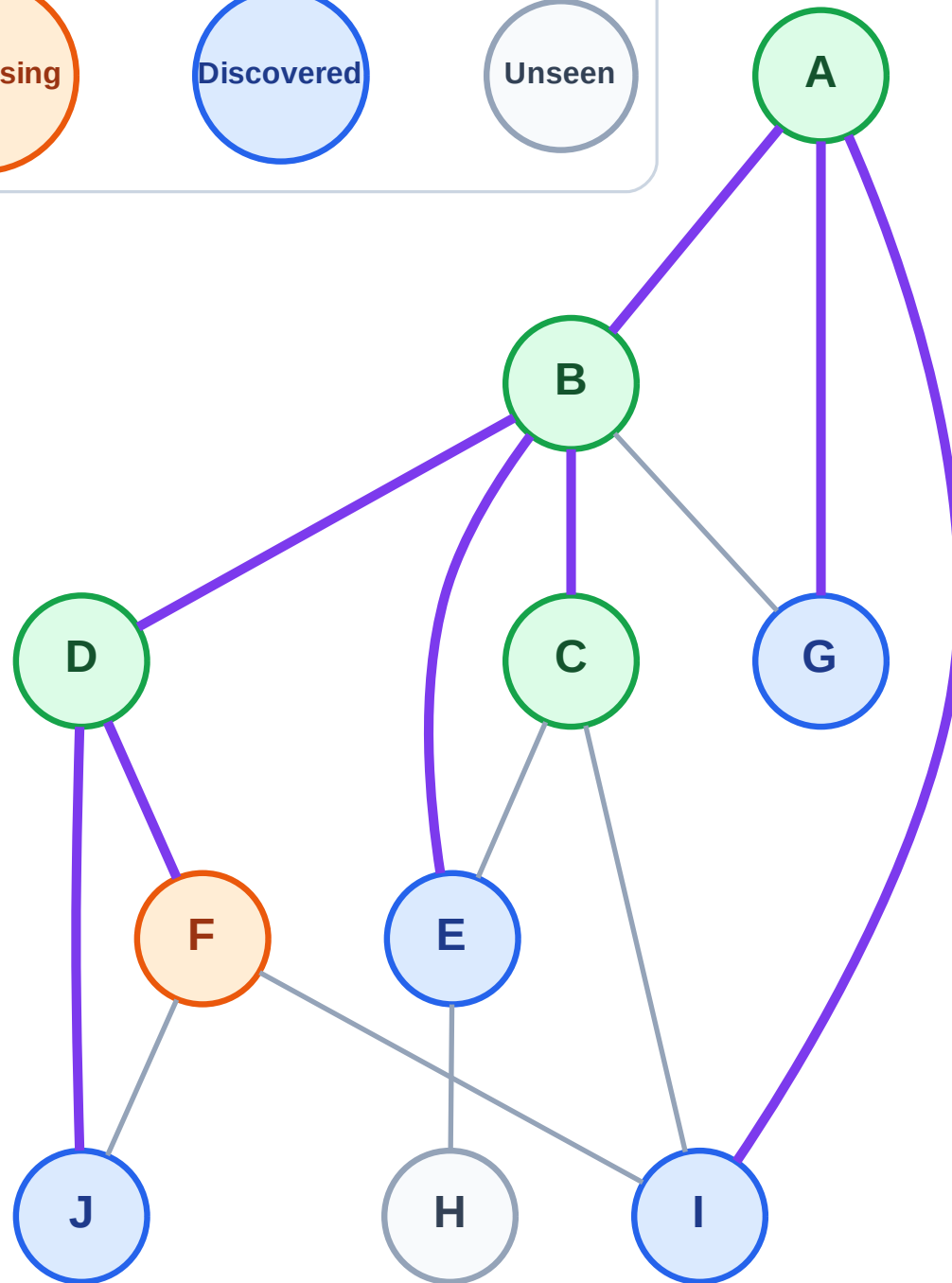
Step 10: Process node F

Legend



Stack (top at right): I | G | E | J

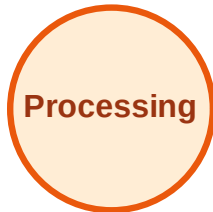
Visited: A, B, C, D



DFS Traversal

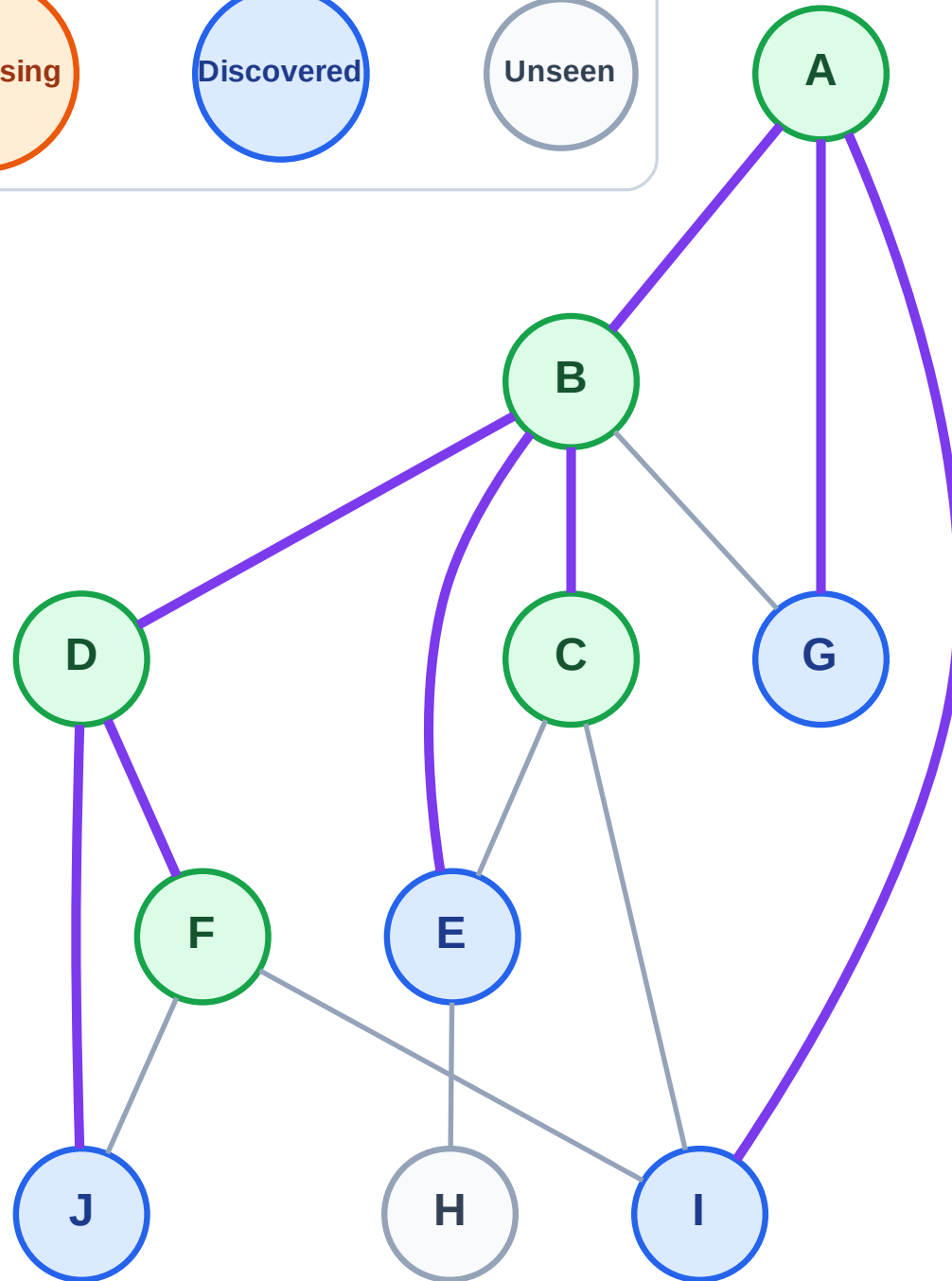
Step 11: No new neighbors from F

Legend



Stack (top at right): I | G | E | J

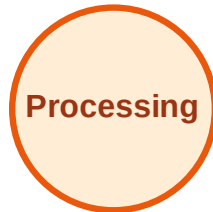
Visited: A, B, C, D, F



DFS Traversal

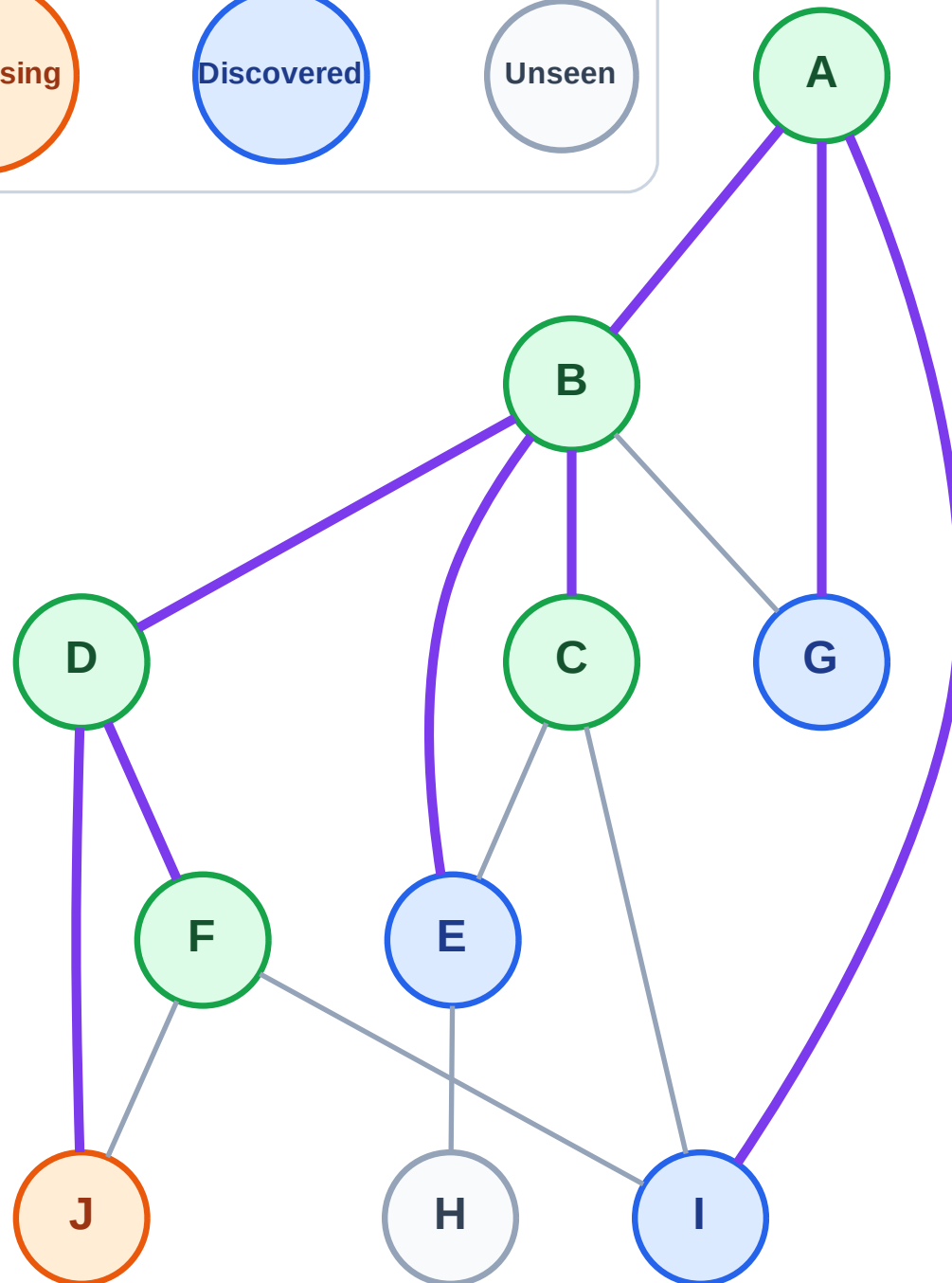
Step 12: Process node J

Legend



Stack (top at right): I | G | E

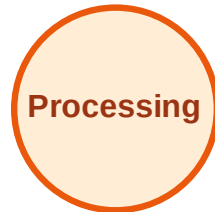
Visited: A, B, C, D, F



DFS Traversal

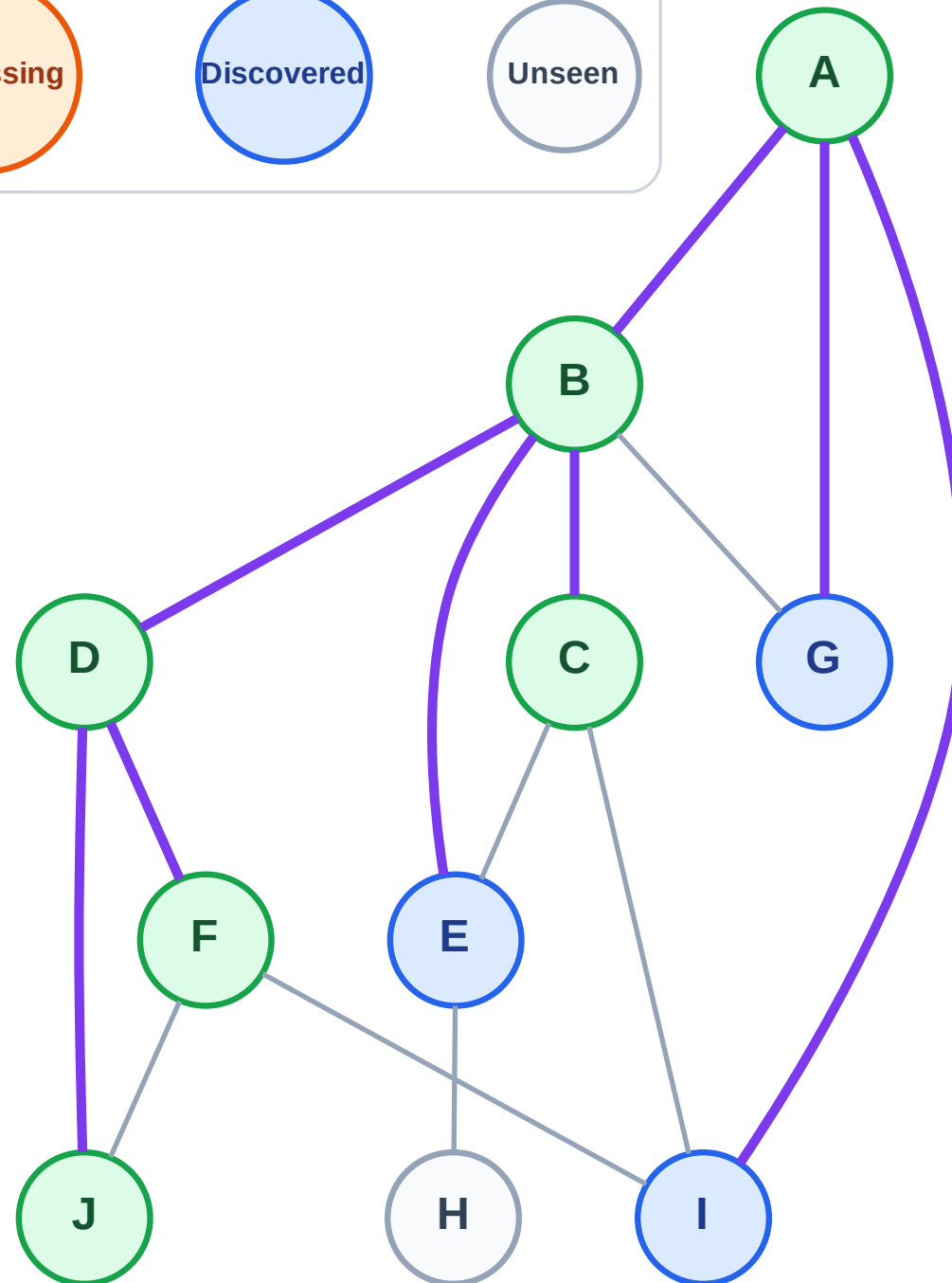
Step 13: No new neighbors from J

Legend



Stack (top at right): I | G | E

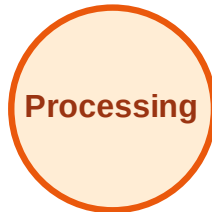
Visited: A, B, C, D, F, J



DFS Traversal

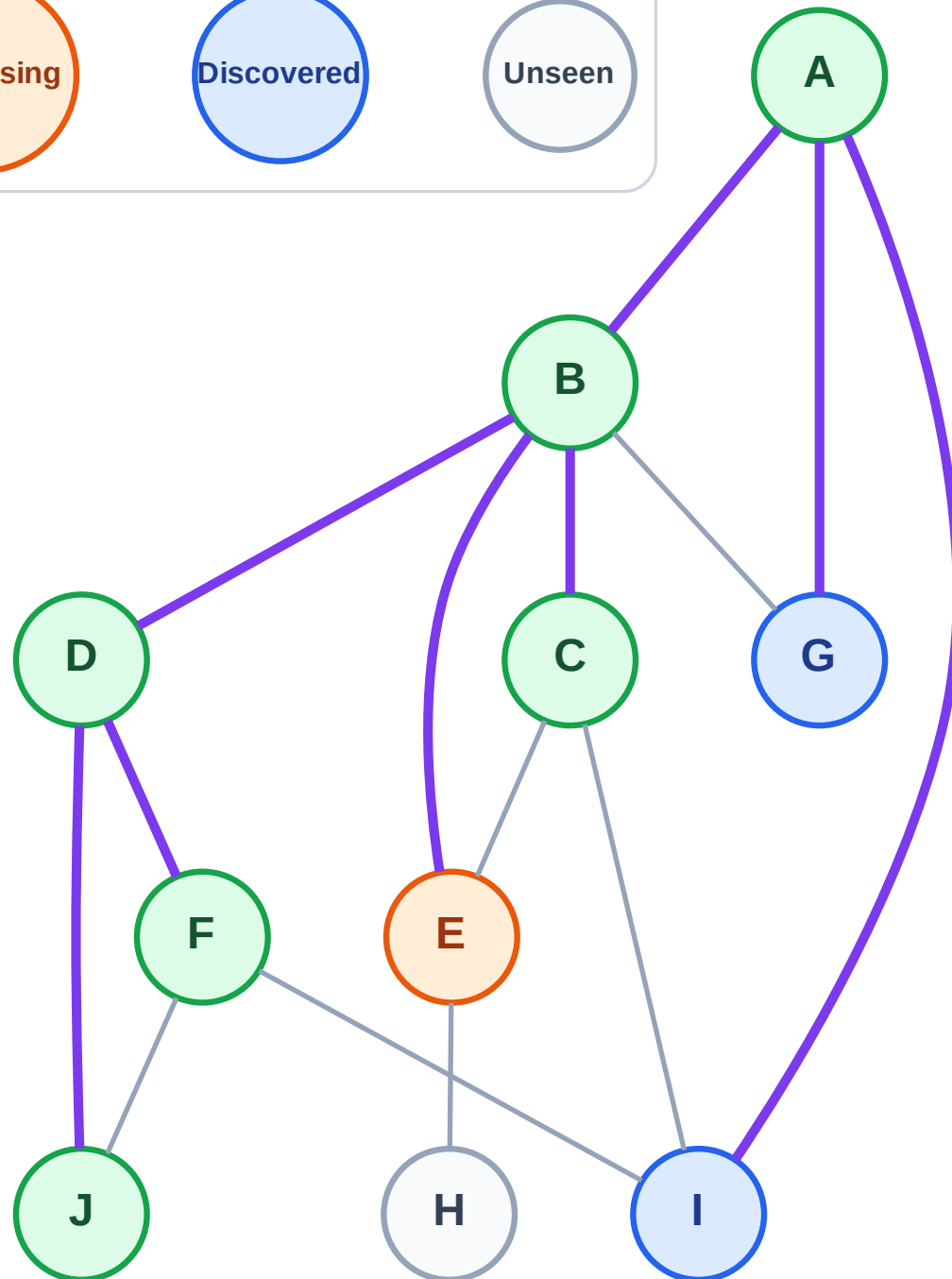
Step 14: Process node E

Legend



Stack (top at right): I | G

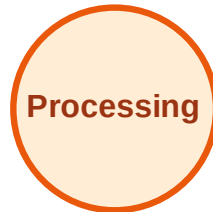
Visited: A, B, C, D, F, J



DFS Traversal

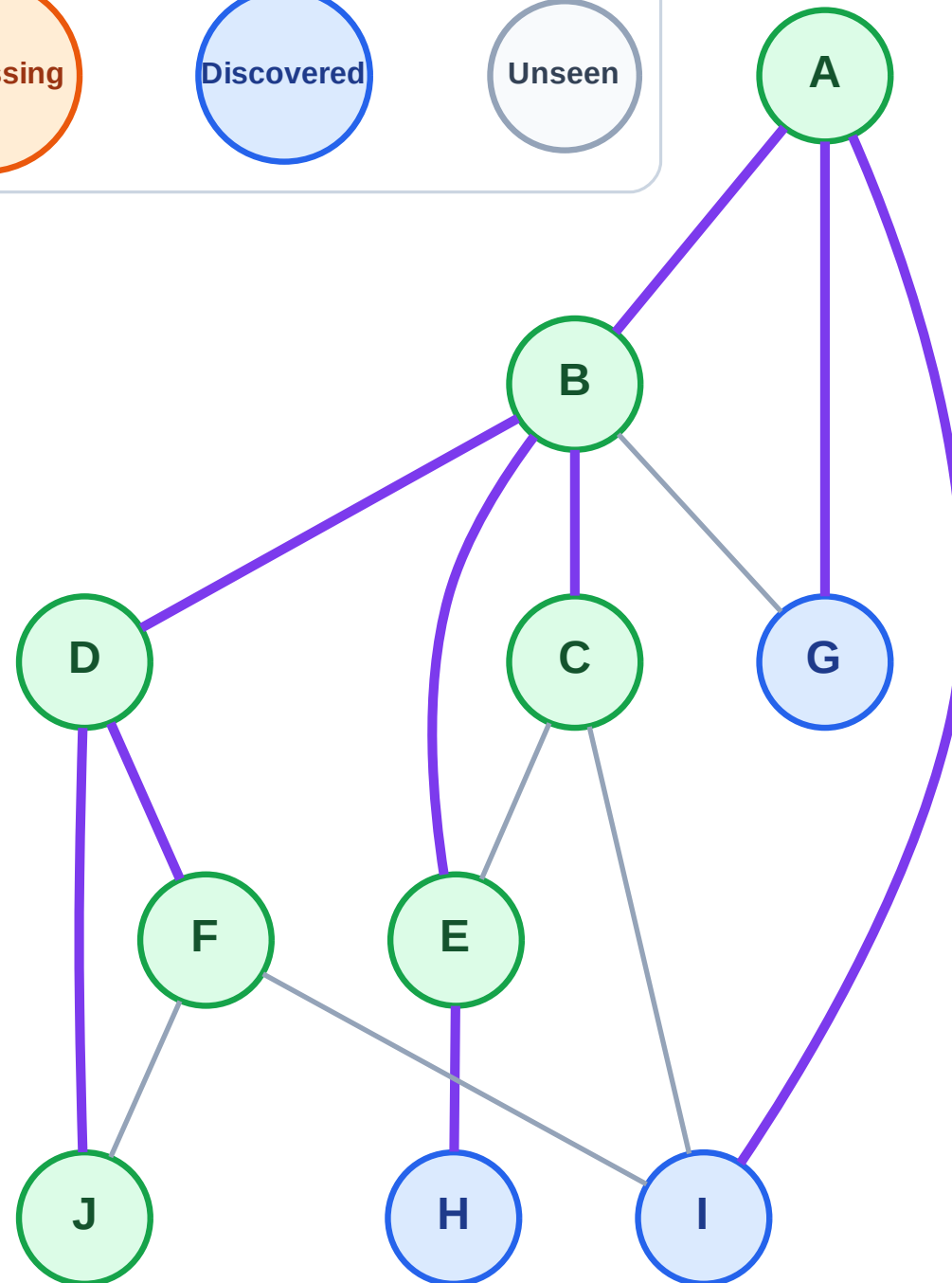
Step 15: Discover H from E

Legend



Stack (top at right): I | G | H

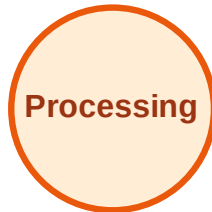
Visited: A, B, C, D, E, F, J



DFS Traversal

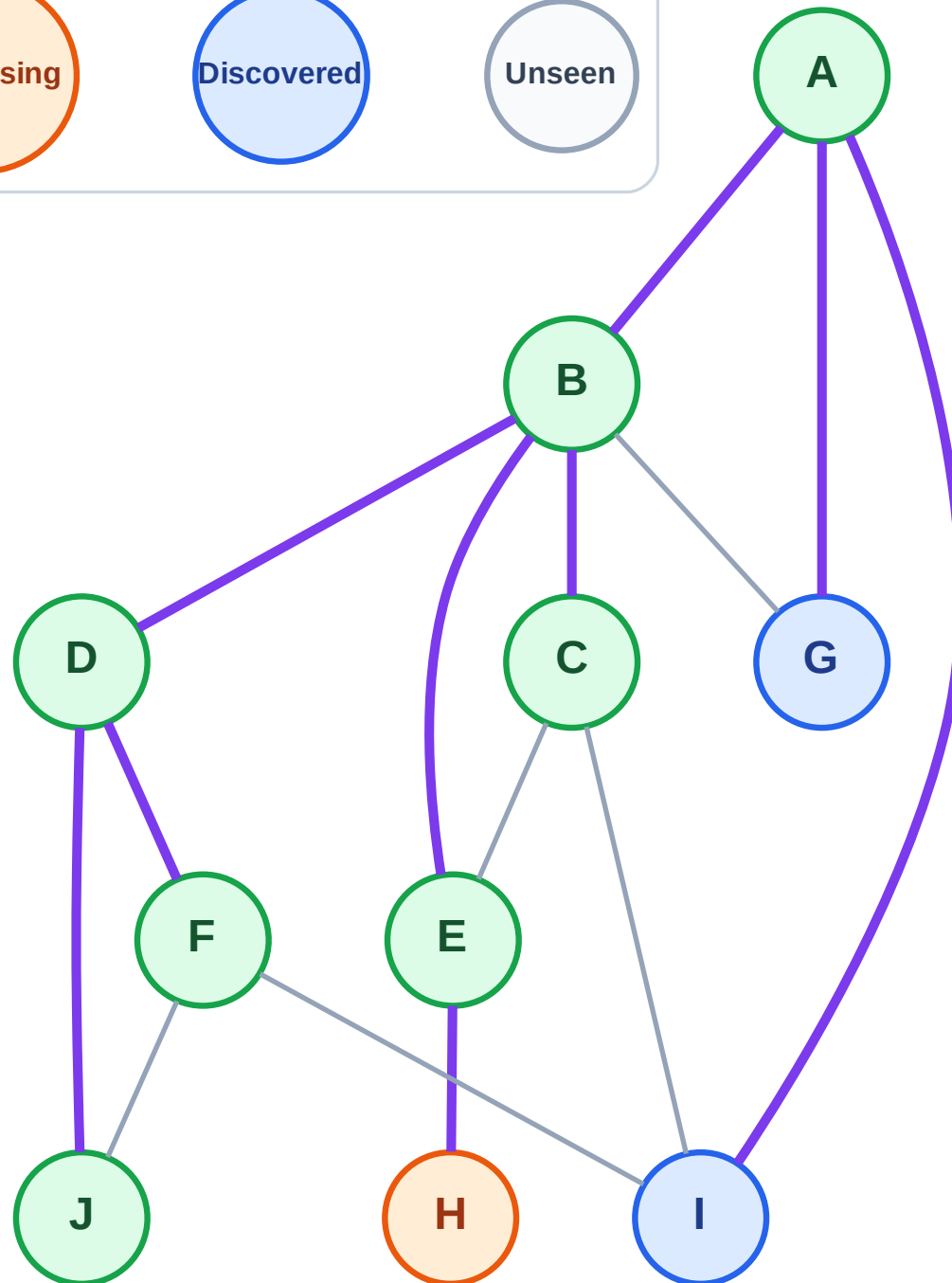
Step 16: Process node H

Legend



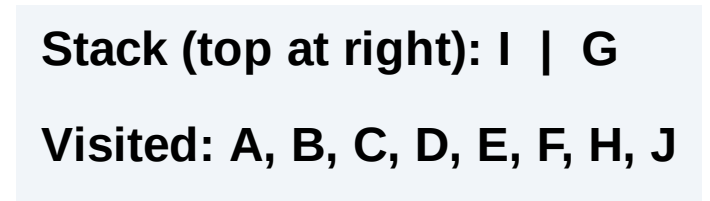
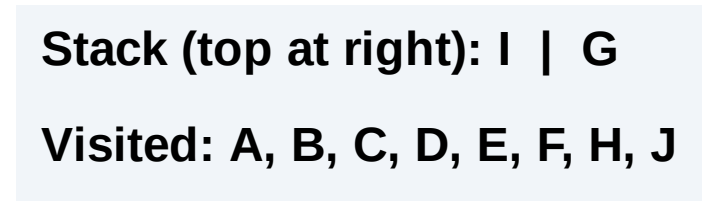
Stack (top at right): I | G

Visited: A, B, C, D, E, F, J



Step 17: No new neighbors from H

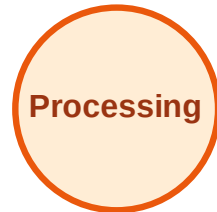
Step 17: No new neighbors from H



DFS Traversal

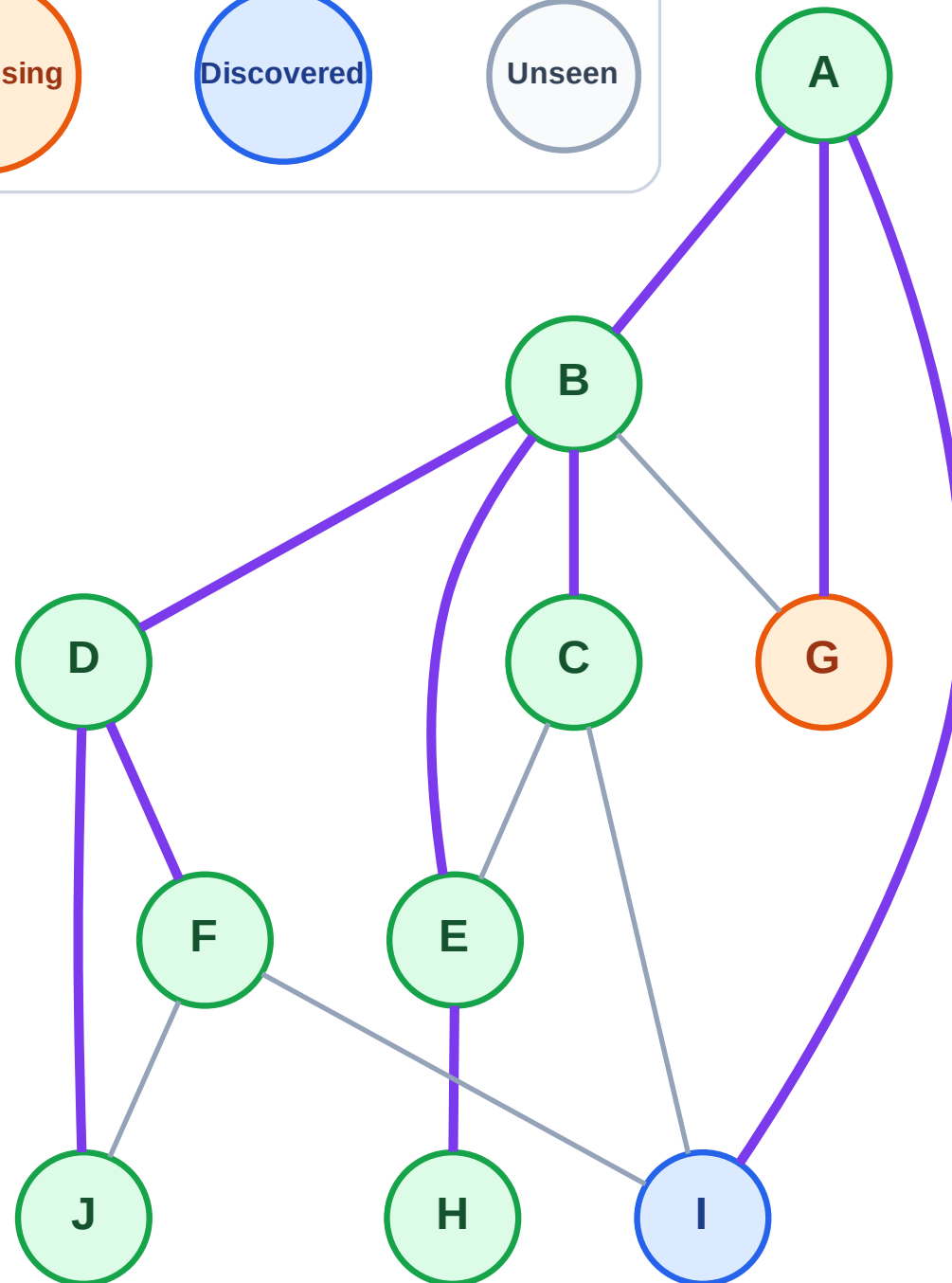
Step 18: Process node G

Legend



Stack (top at right): I

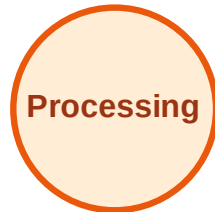
Visited: A, B, C, D, E, F, H, J



DFS Traversal

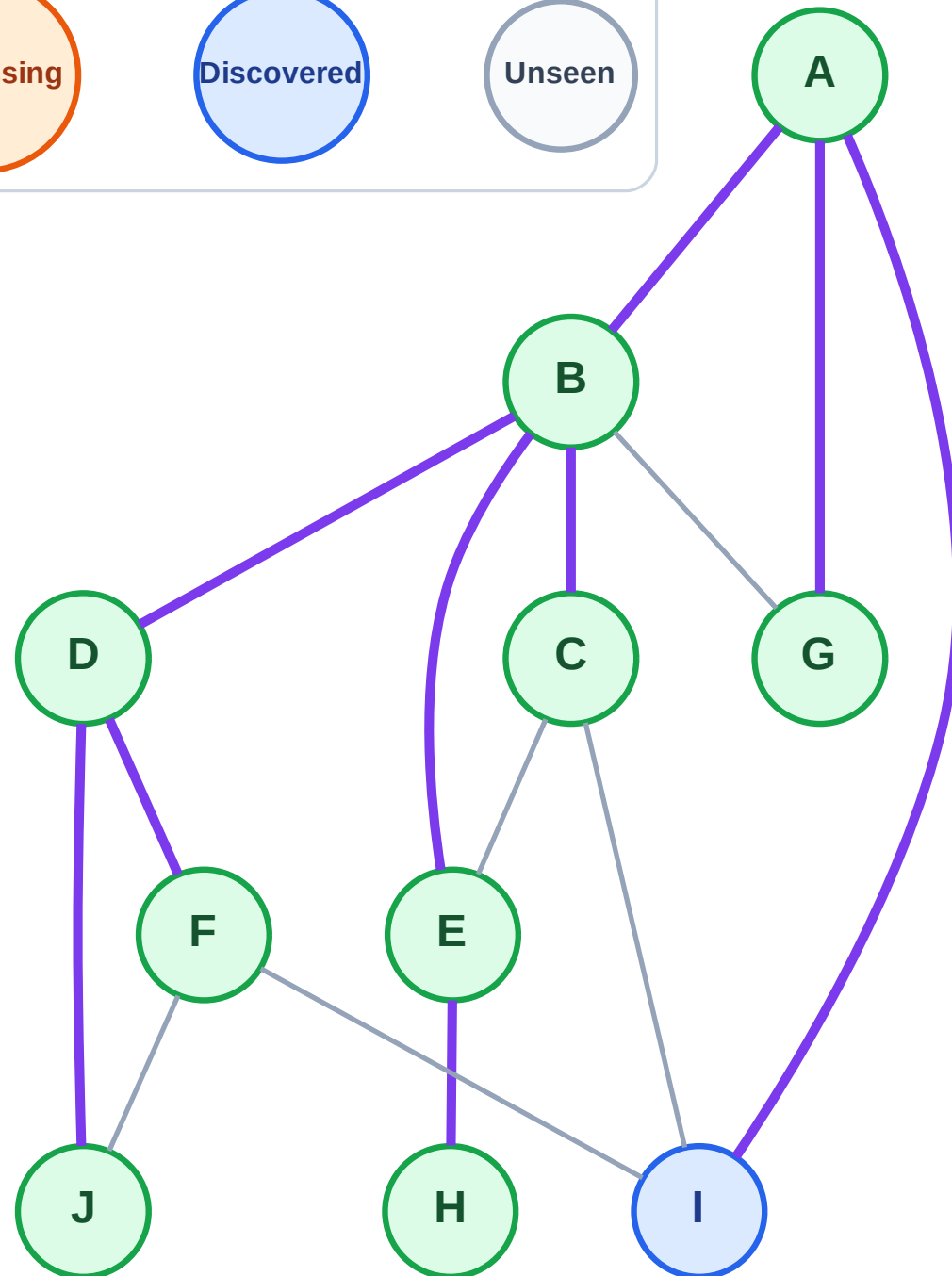
Step 19: No new neighbors from G

Legend



Stack (top at right): I

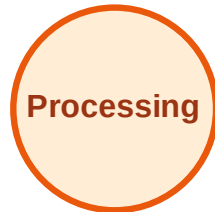
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

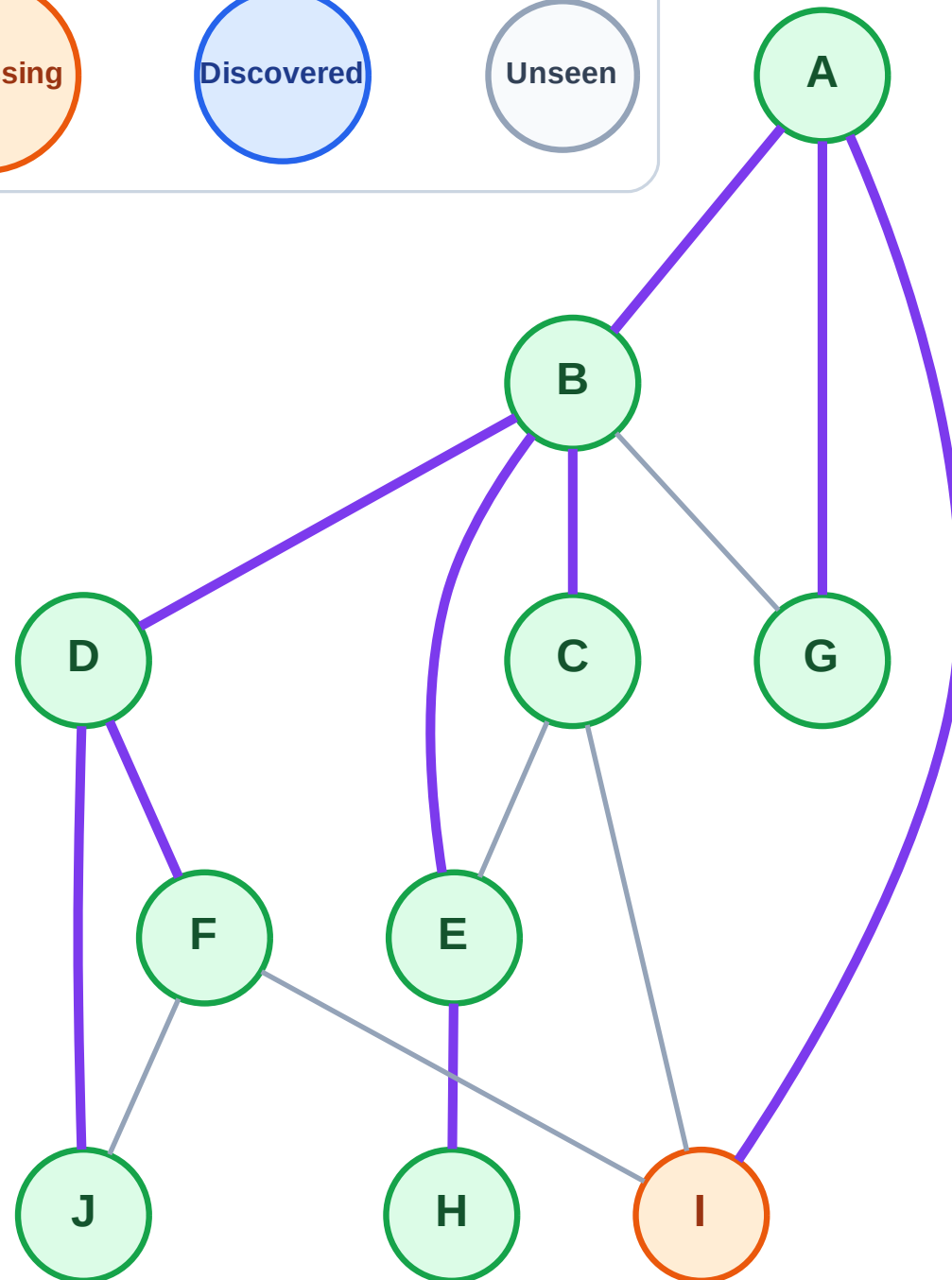
Step 20: Process node I

Legend



Stack (top at right): (empty)

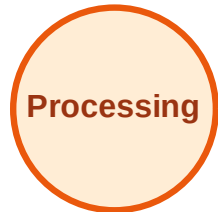
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

